Title
Revealing the Dark Matter of the Metabolome: Libraries, Denoising, and Representation Learning

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Revealing the Dark Matter of the Metabolome:
Libraries, Denoising, and Representation Learning

By

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Abstract

Metabolomics, the comprehensive study of small molecules in biological systems, has become an indispensable tool for modern biomedical research. Unlike genomics and transcriptomics, which capture potential or regulatory information, metabolite profiles reflect the real-time biochemical activities within cells, integrating genetic, enzymatic, and environmental influences. This unique property makes metabolomics particularly powerful for investigating disease mechanisms, nutritional interventions, microbiome interactions, and environmental exposures. At the same time, the concept of the exposome—the dynamic sum of environmental influences such as diet, pollutants, and lifestyle factors—has underscored the urgent need for analytical strategies that can capture both endogenous and exogenous small molecules across diverse biological contexts. High-resolution liquid chromatography–tandem mass spectrometry (LC–MS/MS) has emerged as the principal platform for these efforts, enabling simultaneous detection of thousands of compounds. However, the field continues to face challenges in metabolite annotation, spectral noise reduction, and the development of robust molecular representations for predictive modeling.

This dissertation addresses three major bottlenecks at the interface of metabolomics, lipidomics and exposomics: (1) the incompleteness and heterogeneity of spectral libraries, (2) the prevalence of noise and chimeric spectra that compromise annotation confidence, and (3) the limitations of current molecular machine learning models for property prediction such as chromatographic retention times. To expand chemical space coverage, I developed LibGen, a fully automated pipeline for spectral library generation and curation. LibGen integrates mass error correction, chemistry-aware denoising based on subformula plausibility, and entropy-based quality metrics, allowing reproducible and scalable construction of high-quality libraries. Using this framework, I created the first public ultraviolet photodissociation (UVPD) and electron-activated dissociation (EAD) MS/MS libraries for natural products, demonstrating their superior fragmentation information content compared to conventional collision-induced dissociation. These

resources are openly available and provide the foundation for more confident annotation across diverse chemical classes.

To address the challenge of spectral noise, I introduced a novel Spectral Denoising algorithm that integrates electronic and chemical noise modeling. By leveraging intensity distributions and chemically valid subformula losses, this method distinguishes true fragment ions from spurious peaks, substantially improving sensitivity without sacrificing specificity. Benchmarking in plasma samples revealed a doubling of confident metabolite and exposome annotations compared to conventional noise-threshold approaches, particularly for trace-level compounds at nanomolar or picomolar concentrations. This advance not only improves annotation coverage but also enhances the reliability of exposome-wide association studies.

Finally, to overcome the expressivity limits of conventional message-passing graph neural networks (GNNs), I developed MolRex, a foundation-style molecular representation learning framework. MolRex augments molecular graphs with global connectivity, integrates 3D distance-aware encodings, and fuses graph embeddings with descriptor and fingerprint information using gated mechanisms. The model achieved state-of-the-art performance in predicting chromatographic retention times across multiple LC methods and showed broad applicability in molecular property prediction tasks. This contribution establishes a more generalizable and chemically informed representation for downstream metabolomics and exposomics applications.

Together, these innovations—automated spectral library curation, advanced spectral denoising, and foundation-style molecular embeddings—chart a coherent trajectory for advancing metabolomics toward greater reproducibility, sensitivity, and predictive power. By tackling critical gaps in chemical space coverage, annotation quality, and computational modeling, this work strengthens the analytical and computational foundations of exposome science. Ultimately, the approaches developed here aim to accelerate discovery in metabolomics by providing more reliable tools for identifying small molecules and by enabling deeper integration of genetic, biochemical, and environmental data in human health research.

Introduction

Metabolomics—the comprehensive analysis of small molecules in biological systems—has become a cornerstone of modern biology, as it connects molecular processes to physiological outcomes[1-3]. Metabolites represent the downstream products of gene expression, enzymatic activity, and environmental interactions[4-6]. Their abundances and chemical forms reflect not only genetic control but also the influence of diet, microbiota, and environmental exposures[7, 8]. Unlike the genome or transcriptome, which indicate potential biological states, the metabolome captures the realized biochemical phenotype—a direct snapshot of metabolic activity occurring in cells and organisms at a given moment[9, 10]. This integrative perspective makes metabolomics uniquely suited to reveal functional pathways underlying health and disease.

Over the past two decades, metabolomics has generated transformative insights across nearly every domain of human science and disease, offering mechanistic depth that complements genomic and proteomic data. In cancer biology, metabolic profiling has revealed the profound reprogramming of cellular metabolism that supports rapid proliferation and survival[11-13]. Studies have shown how tumors shift from oxidative phosphorylation to aerobic glycolysis (the Warburg effect) and enhance glutaminolysis to fuel nucleotide and lipid biosynthesis[14, 15]. Metabolomics was crucial in the discovery of oncometabolites such as 2-hydroxyglutarate, succinate, and fumarate, whose accumulation in specific genetic contexts (e.g., IDH or SDH mutations) alters DNA and histone methylation, directly linking metabolism to epigenetic regulation and tumorigenesis[16-19]. These findings would have remained obscured without direct metabolite-level measurements. In cardiometabolic disorders, metabolomics has elucidated how systemic metabolic dysregulation precedes disease onset and provides actionable biomarkers for early intervention. Elevated plasma levels of branched-chain amino acids, acylcarnitines, and the gut microbial metabolite trimethylamine-N-oxide (TMAO) have been associated with insulin resistance, mitochondrial dysfunction, and atherosclerotic risk[20-24]. Unlike traditional biomarkers that merely correlate with disease, these metabolites revealed causal biochemical pathways, identifying the role of the gut microbiome and hepatic flavin monooxygenases in generating TMAO and establishing new therapeutic targets[25]. Such examples illustrate how metabolomics enables "reverse translation," where statistically derived molecular patterns guide mechanistic experiments. Neurological research has also benefited immensely from metabolomic technologies. Profiling of cerebrospinal fluid and brain tissue has revealed alterations in tryptophan–kynurenine metabolism, oxidative stress markers, and acylcarnitine profiles in disorders such as Alzheimer's, Parkinson's, and Huntington's disease[26-29]. These results have shed light on impaired mitochondrial energy metabolism and neuroinflammation, suggesting new avenues for intervention that go beyond protein aggregation models. More recently, real-time metabolic imaging and stable isotope tracing have begun to map neurotransmitter fluxes in vivo, bridging metabolic pathways with neural circuit function[30, 31]. Beyond disease-specific applications, metabolomics has transformed understanding of host–microbiome–environment interactions. Studies integrating microbial genomics with metabolite profiling have shown that gut bacteria produce a wide range of bioactive molecules—including short-chain fatty acids, secondary bile acids, and indole derivatives—that regulate host immunity, metabolism, and even neurobehavioral processes[32, 33]. Collectively, these findings highlight that metabolites function as molecular messengers connecting microbial and host biology, translating chemical signals across kingdoms to regulate immunity, metabolism, and behavior.

The growth of metabolomics has been tightly coupled to advances in analytical chemistry, particularly mass spectrometry and chromatography. Among the available tools, liquid chromatography (LC) coupled with high-resolution tandem mass spectrometry (MS/MS) has become the dominant platform for untargeted metabolomics[34, 35]. Chromatographic separation mitigates ion suppression, resolves structural isomers, and provides retention time as an orthogonal identifier, while high-resolution MS delivers accurate mass and isotopic fidelity for molecular formula assignment[36]. Fragmentation spectra (MS/MS) offer rich structural clues that enable both targeted and untargeted identification. By comparing experimental MS/MS spectra with reference spectral library, researchers are able to identify key biomarkers with high confidence. Together, these features allow LC–MS to detect thousands of metabolites spanning a wide range of polarity, molecular weight, and concentration—making it uniquely capable of comprehensive coverage across complex biological matrices. On the other hand, gas chromatography–mass spectrometry (GC–MS) remains a gold standard for small, volatile, and thermally stable compounds, offering excellent chromatographic resolution, retention-time reproducibility, and extensive spectral libraries such as NIST and FiehnLib[37-39]. However, its requirement for chemical derivatization complicates sample preparation, can induce degradation or artifact formation, and restricts detection to metabolites amenable to volatilization. Consequently, GC–MS covers only a narrow chemical space compared to LC–MS, missing most lipids, peptides, and other nonvolatile compounds central to biological regulation[39]. Capillary electrophoresis–mass spectrometry (CE–MS) offers complementary strength in separating ionic and highly polar species such as amino acids, nucleotides, and organic acids, often with sub-minute resolution[40, 41]. Yet CE–MS suffers from low robustness, sensitivity to minor changes in buffer composition, and migration-time variability that complicates alignment across batches. Its inherently small injection volume also limits sensitivity for low-abundance metabolites. Compared to LC–MS, CE–MS provides higher separation efficiency but lower reproducibility, narrower dynamic range, and poorer compatibility with hydrophobic molecules, which together hinder its scalability for large cohort studies[42]. Nuclear magnetic resonance (NMR) spectroscopy, in contrast, occupies a distinct niche emphasizing quantitative precision and structural confidence. NMR can unambiguously identify metabolites, distinguish structural isomers, and quantify absolute concentrations without external calibration, as it is particularly valuable for stable-isotope tracing and metabolic flux analysis[43-45]. Nonetheless, its detection limit typically lies in the high micromolar range—several orders of magnitude less sensitive than LC–MS—and its instrument cost and spectral complexity restrict coverage to fewer than a hundred abundant metabolites per sample[46]. As such, NMR excels in quantification and reproducibility but lacks the depth of coverage and sensitivity required for systems-scale metabolomics. Direct-infusion mass spectrometry (DIMS) enables ultra-fast metabolite profiling without chromatographic separation, greatly increasing throughput but at the cost of severe ion suppression and ambiguous spectral overlap, making it only suitable for generating mass spectral libraries, but not good analytical approach for analyzing biological samples due to its difficulty in making high confident annotation[47, 48]. Mass spectrometry imaging (MSI) techniques such as MALDI-MSI and DESI-MSI visualize the spatial distribution of metabolites within tissues, providing valuable anatomical context[49]. However, they typically lack chromatographic separation, suffer from lower mass accuracy and reproducibility, and are limited in chemical annotation compared to LC–MS workflows[50]. As each analytical platform contributes unique advantages but also inherent limitations in sensitivity, chemical coverage, or reproducibility, LC–MS/MS has therefore emerged as the most balanced and scalable solution for both targeted and untargeted metabolomics, offering broad molecular coverage, high sensitivity, and compatibility with diverse biological and environmental matrices. Its ability to integrate quantitative precision with structural specificity has made LC–MS the central analytical engine of modern metabolomics and the foundation for most subsequent computational developments in this field. Altogether, these complementary methods collectively define the analytical foundation of metabolomics, with LC–MS/MS serving as the most versatile compromise between coverage, sensitivity, and throughput.

As the analytical scope of metabolomics expands, its intersection with environmental health sciences has crystallized through the concept of the exposome—the totality of environmental exposures that influence biology across the lifespan. Since its introduction by Christopher Wild in 2005, the exposome framework has emphasized that genetics alone cannot explain most chronic diseases: environmental and lifestyle factors are estimated to account for more than 70% of disease burden worldwide[51]. Capturing these influences experimentally, however, is extraordinarily challenging because exposures are chemically diverse, transient, and occur at trace concentrations[52-54]. High-resolution LC–MS/MS has emerged as the only practical means to measure the exposome directly, detecting both endogenous metabolites and exogenous compounds such as pollutants, drugs, dietary chemicals, and their transformation products[55]. Large-scale exposome studies have revealed hundreds of previously unrecognized xenobiotics in human plasma and urine, demonstrating how pervasive environmental chemicals are in daily life[56, 57]. These discoveries have reshaped public-health paradigms by linking specific exposures, such as pesticide metabolites, plasticizers, and flame retardants, to metabolic dysregulation and disease onset[58, 59].

Metabolomics thus provides the analytical engine for exposome science, allowing environmental chemistry and molecular biology to converge into a single, quantifiable framework.

Despite its success, metabolomics still lags behind genomics and proteomics in standardization, reproducibility, and computational maturity[60]. Whereas the Human Genome Project established a complete genetic reference and proteomics now benefits from well-defined peptide spectral libraries and deterministic fragmentation rules[61, 62], metabolomics lacks any comparable foundation. Each small molecule exhibits distinct ionization behavior and fragmentation chemistry that vary across instruments, collision energies, and adduct forms. Existing spectral libraries such as GNPS, NIST, METLIN, and MassBank.us[38, 63-65], although they collectively host millions of reference MS/MS spectra, still cover less than two percent of known chemical space and remain heavily biased toward drug-like molecules[66, 67]. Entire classes of biologically important compounds, including dietary phytochemicals, microbial metabolites, and exposome compounds, remain underrepresented. Without comprehensive reference data, annotation remains incomplete and inconsistent across laboratories. The challenge is further worsened by the problem of noise ions in MS/MS spectra. At the low concentrations typical of exposome studies, MS/MS spectra are often contaminated heavily by electronic and chemical noise, generating chimeric spectra in which genuine fragment ions are buried among spurious peaks[52, 68, 69], leading to the large number of unannotated spectra in metabolomics and exposome studies. Unlike peptides, which fragment along predictable amide bonds allowing algorithmic filtering, small molecules follow no universal fragmentation grammar. Traditional denoising methods rely on intensity-thresholding, which may discard informative low-abundance fragments or retain implausible ions, undermining annotation accuracy[70]. As a result, reliable spectral denoising methods still remain major bottlenecks in metabolomics workflows.

From a computational standpoint, the representation learning revolution that transformed genomics and proteomics has not yet reached full potential in metabolomics[71]. Deep learning models such as AlphaFold, ESM, and ProtT5 have achieved remarkable accuracy in predicting protein structures and functions because biological macromolecules are governed by linear sequence grammars and well-defined alphabets[72-74].

Small molecules, by contrast, are non-templated, graph-structured entities with immense structural diversity and context-dependent properties. Conventional message-passing graph neural networks (GNNs) capture only local chemical neighborhoods and struggle with long-range interactions like aromatic stacking, hydrogen bonding, or 3D conformational strain, factors that critically influence chromatographic retention, solubility, and bioactivity[75, 76]. Moreover, physicochemical behavior depends strongly on experimental context: the same molecule can exhibit different ionization states or adduct patterns under varying solvent and instrument conditions[77, 78]. This context dependence renders pre-trained molecular embeddings less transferable than their protein or genomic counterparts.

One of the core knots of these analytical and computational issues is the open-set nature of the metabolome. Unlike genomics or proteomics, which operate within finite alphabets of nucleotides or amino acids, metabolomics explores an effectively boundless chemical universe[79]. Novel metabolites, transformation products, and adduct species continue to be discovered, and no exhaustive enumeration of molecular space is possible. The lack of deterministic fragmentation pathways and the influence of acquisition parameters mean that reproducibility depends as much on experimental standardization as on computational interpretation. Consequently, metabolomics faces a fundamental challenge: it must function without ever achieving a closed reference set. These realities explain why the strategies that revolutionized genomics and proteomics (universal sequence references, deterministic fragmentation rules, and sequence-based deep learning) cannot simply be transplanted to metabolomics. Instead, the field requires new paradigms tailored to small-molecule complexity, integrating chemical knowledge, data-driven inference, and scalable automation.

To this end, this dissertation addresses these challenges through three synergistic developments that collectively advance the analytical and computational foundations of metabolomics. First, an automated spectral library generation pipeline, LibGen, was developed to curate, standardize, and denoise reference MS/MS spectra at scale. LibGen corrects mass errors, filters noise ions, and evaluates spectral quality using the proportion of true fragment ions, enabling the rapid construction of high-quality reference libraries.

This framework not only facilitates reliable compound annotation in current workflows but also lays the groundwork for developing computational tools analogous to those that transformed proteomics and genomics. Second, a chemistry-aware algorithm, *Spectral Denoising*, was designed to distinguish genuine fragment ions from spurious peaks based on subformula plausibility rather than arbitrary intensity thresholds. This approach enhances sensitivity toward low-abundance metabolites and improves spectral quality. When integrated with entropy-based spectral similarity search, we developed this *Denoising Search*, which markedly increases the number of annotatable compounds, recovering meaningful signals from data previously regarded as "dark matter" in metabolomics, and substantially expanding the accessible chemical space. Third, a foundation-style molecular representation learning framework, MolRex, was introduced to integrate graph-based molecular topology with domain specific features including descriptors and fingerprints. Combining with novel GNN architecture and augmented graph, MolRex was able to achieve global receptive filed for any GNN backbone but preserve the original structure with novel chemistry-aware edge encoding. By capturing both local bonding environments and global structural context, MolRex enables robust prediction of molecular properties—including chromatographic retention time—across heterogeneous datasets. The predicted retention time demonstrates strong generalizability across laboratories and chromatographic systems, providing orthogonal dimension of information that directly enhances the confidence of MS/MS-based compound annotation. Together, these developments establish a unified computational–analytical framework that enhances data quality, interpretability, and generalizability in metabolomics—bringing the field closer to the scalability and reproducibility long realized in genomics and proteomics.

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Chapter 1: LibGen: generating high quality spectral libraries of natural products for EAD-, UVPD- and HCD-high resolution mass spectrometers

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Abstract

Compound annotation using spectral-matching algorithms is vital for (MS/MS)-based metabolomics research, but hindered by the lack of high-quality reference MS/MS library spectra. Finding and removing errors from libraries, including noise ions, is mostly done manually. This process is both error-prone and time-consuming. To address these challenges, we have developed an automated library curation pipeline, LibGen, to universally build novel spectral libraries. This pipeline corrects mass errors, denoises spectra by subformula assignments, and performs quality control of the reference spectra by calculating explained intensity and spectral entropy. We employed LibGen to generate three high-quality libraries chemical standards of 2,241 natural products. To this end, we used an IQ-X orbital ion trap mass spectrometer to generate 1,947 classic high-energy collision dissociation spectra (HCD) as well as 1,093 ultraviolet-photodissociation (UVPD) mass spectra. The third library was generated by an electron-activated collision dissociation (EAD) 7600 ZenoTOF mass spectrometer yielding 3,244 MS/MS spectra. The natural compounds covered 140 chemical classes from prenol lipids to benzypyrans with >97% of the compounds showing <0.2 Tanimoto-similarity, demonstrating a very high structural variance. Mass spectra showed

much higher information content for both UVPD- and EAD-mass spectra compared to classic HCD spectra when using spectral entropy calculations. We validated the denoising algorithm by acquiring MS/MS spectra at high-concentration and at 13-fold diluted chemical standards. At low concentrations, a higher proportion of spectra showed apparent fragment ions that could not be explained by subformula losses of the parent molecule. When more than 10% of the total intensity of MS/MS fragments was regarded as noise ions, spectra were considered as low quality and were not included in the libraries. As the overall process is fully automated, LibGen can be utilized by all researchers who create or curate mass spectral libraries. The libraries we created here are publicly available at MassBank.us.

1. Introduction

Liquid chromatography coupled with high-resolution tandem mass spectrometry (LC-MS/MS) is the leading tool in metabolomic studies for measuring small organic molecules in biological systems[1,2]. However, the metabolome is constituted of extremely diverse chemicals, unlike genes or proteins[3, 4]. Mass spectral-based annotation of metabolites and exposome chemicals rely on collections of reference spectra acquired on a variety of LC-MS/MS systems and conditions[5, 6]. The larger and the more diverse such spectral libraries are, the better the chances to identify unknowns in metabolomic screens. While current public and licensed libraries consist of millions of mass spectra, the total number of molecules is less than 2% of PubChem's 113 million compounds [7, 8, 9, 10]. Moreover, many molecules give scarce mass spectra with few fragment ions, leading to high false discovery rates to disambiguate isomers, for example, for lipid double bond positions. Here, we used new mass spectral fragmentation mechanisms, ultraviolet photodissociation (UVPD) and electron activated dissociation (EAD), to yield more informative and more exhaustive MS/MS spectra[11, 12].

Creating high-quality mass spectral libraries is not straightforward. Spectra from small molecules must account for a variety of in addition to in-source fragmentation, isobaric interferences, satellite ions due to Fourier transformation processes[13], electronic noise[14] or unstable instrument conditions[15]. Typically, users create libraries by a discarding low-intense fragment ions and manually curating data to ensure high-quality mass spectra[16]. For example, the NIST MS interpreter tool is used for manual curation of NIST libraries[17] but cannot be used in batch mode without human interaction. The batch-mode data curation tool Curatr[16] does not perform any quality control. SIRIUS and MS-FINDER show decent performance on annotating fragment peaks with neutral losses but have poor tolerance for recognizing a large number of radical ion losses in mass spectra[18]. While Rmassbank[14] employs spectra cleaning steps, it requires one spectrum per file which disables the use of multi-compound LC-MS/MS runs. We here present LibGen, an automatic tool to curate and quality-control mass spectra files from analyses of mixtures of chemical compounds, and apply this tool to generate libraries of new MS/MS fragmentation mechanisms using an improved denoising algorithm based on subformula assignments.

2. Methods

Acquiring tandem mass spectral data for MS/MS libraries

For all libraries, authentic standards were prepared in mixtures of 25 non-isomeric compounds each at 1 mg/L in methanol and stored at -20°C. (1) EAD spectra were acquired using an Exion LC AD liquid chromatograph system (SCIEX, ON, Canada) coupled with a Sciex ZenoTOF 7600 quadrupole-time of flight (QTOF) mass spectrometer (Sciex, ON, Canada). Chromatographic separation was performed with a reversed-phase, 30 mm length x 2.1 mm inner diameter (i.d.) Kinetex pentafluorophenyl-based (PFP) column with 1.7-µm particle size (Phenomenex Inc, Torrance, CA), equipped with a 2.1 mm i.d. SecurityGuard™ ULTRA Cartridges precolumn (Phenomenex Inc, Torrance, CA). Flow rate was 0.8 ml/min on a water/acetonitrile gradient with 4 min data acquisitions from t_0 to t_{max}. MS/MS spectra were acquired by creating a precursor ion inclusion list comprising $[M+H]^+$, $[M+Na]^+$, and $[M+NH_4]^+$ adduct forms for each standard in each mixture using information-dependent analysis (IDA) mode with 60 ms EAD reaction time. No dynamic exclusion window was used to ensure we captured multiple EAD-CID MS/MS spectra per reference standard, including isomers. Fragmentation was achieved with 12 eV EAD and 30 eV CID in series for each individual MS/MS spectrum. The Zeno trap was active for all MS/MS scans. All spectra were stored in centroid mode. (2) Both UVPD and HCD spectra were acquired on a ThermoFisher Orbitrap IQ-X tribrid mass spectrometer coupled with a Vanquish UHPLC system (ThermoFisher Scientific, San Jose, CA). Chromatographic separation was performed with a reversed-phase 130Å, 1.7 µm particle-size, 100 mm length x 2.1 mm i.d. Waters bridged ethylene hybrid C18 column (Waters, Milford, MA). Flow rate was 0.6 ml/min on a water/acetonitrile gradient with 10 min data acquisitions from t_0 to t_{max}. Electrospray was performed at 3.5 kV in positive mode and 2.5 kV in negative mode at 275 ºC capillary temperature. MS/MS spectra were acquired by creating a precursor ion inclusion list comprising $[M+H]^+$, $[M+NH_4]^+$, $[M-H]^-$, and $[M+CH_3COO]^-$ adduct forms for each standard in each mixture. In addition, top-4 data-dependent MS/MS spectra were acquired by HCD at 30 NCE and HCD+UVPD fragmentation with HCD at 30 NCE concomitant with UVPD fragmentation using a 213 nm laser with 200 msec activation time. All spectra were acquired in profile mode and subsequently centroided using MSConvert[19].

Implementing LibGen

Python v3.8 was used to create the LibGen framework. The first 14 characters of InChI keys provided by vendors were validated against PubChem[7] and GNPS[8] databases to retrieve SMILES codes of achiral (2D) structures. SMILES codes were used to calculate accurate masses using RDKit[20]. Chemical class information was acquired using ClassyFire and stored as metadata[21]. Monoisotopic mass of precursors were calculated using an adduct calculator[22]. Spectral entropy and entropy similarity were computed with the flash-entropy module[23]. Raw spectra files were converted from instrument files to 64-bit mzML files using MSConvert in Proteowizard[19]. Distinct functionalities in Libgen are realized through discrete modules, promoting greater efficiency and facilitating code reusability. This modular architecture of Libgen provides advantages for both novice users seeking a simple all-in-one mode for high-throughput processing of mass spectrometry libraries, as well as experienced users who wish to tailor their workflows to meet specific requirements. Precursor ions from MS/MS spectra were removed, and fragment ions were binned to 0.020

Da windows to match the R=15,000 mass spectral resolving power of the instruments. Fragment ion intensities were sum-normalized to unity.

3. Results and Discussion

LibGen pipeline to curate MS libraries from reference standards

Natural product libraries were acquired and curated: for electron activated dissociation (EAD), 1,614 reference standards were injected in non-isobaric mixtures into a 7600 ZenoTOF mass spectrometer in positive ESI mode. For both ultraviolet photodissociation (UVPD) and high collision energy (HCD), an Orbitrap IQ-X tribrid mass spectrometer was used, analyzing 2,007 natural products in positive and negative ESI mode. Because UVPD and HCD spectra were obtained using an identical liquid chromatography system and reference standards, a similar number of MS1 features and statistics were derived from these two instruments. The curation process employed in LibGen is depicted in Figure 1. Following the preprocessing of raw spectral data and conversion into feature tables containing peak information, the curation process commenced by matching precursor masses against theoretical precursor mass values in a pre-established standard list, using a mass tolerance of 5 mDa. The curation statistics are summarized in Table 1. All three datasets yielded a match rate of approximately 85% to 88% for injected chemicals. Missed detection of chemical standards was likely due to failure to acquire MS/MS spectra due to poor ionization efficiency that led to low signal/noise in MS1 or to poor quality MS/MS spectra. For ZenoTOF-EAD data acquisitions, chemicals may also have been missed if these were only ionizable in negative ESI mode, not in positive ESI.

LibGen was developed as an out-of-box solution for curating high-quality mass spectral libraries in a fully automatic manner. LibGen simplifies the input process by requiring only a list of InChIKeys[24] for injected chemical standards pointing to specific chromatogram file labels, and resulting raw mass spectra data files. LibGen uses SMILES codes retrieved from InChI keys to neutralize molecules if applicable, followed by recalculation of chemical information from updated SMILES. The prepared standard list can also be used as inclusion list for DDA data collection for minimizing human input error. At current, LibGen uses $[M+H]^+$,

$[M+Na]^+$, $[M+NH_4]^+$, $[M-H_2O+H]^+$ adducts for calculating cation precursors, and $[M-H]^-$, $[M+acetate]^-$, $[M-H_2O-H]^-$, $[M+formate]^-$, $[M+Cl]^-$, $[M+Na-2H]^-$ for anions. Additional adducts can be incorporated by modifying the source code. For each injection mixture, the custom-built feature finding module, ff_droup, generates a single peak matrix to pair retention time, and precursor m/z information with corresponding MS/MS spectra. The detailed description of the ff_droup module is given in Supplement Method 1. As precursor ion abundance in MS/MS spectra depends on instrument conditions and dissociation energies, the residual abundance of precursor ions in MS/MS spectra may obfuscate the orthogonal information of fragmentation ions in MS/MS similarity scoring. We therefore removed precursor ions from MS/MS spectra.

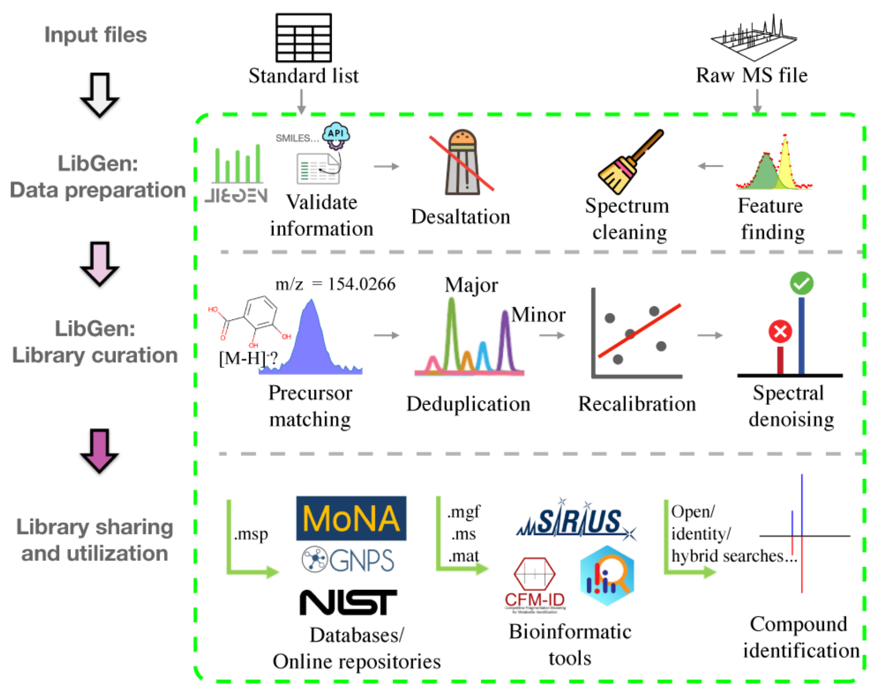

Figure 1. Workflow for curating high-quality mass spectral libraries in LibGen. The software needs a list of target chemicals with corresponding raw mass spectral data. LibGen performs all curation processes and exports a curated library into various formats for upload to MS/MS libraries, downstream bioinformatics tools or advanced compound identification software.

Relying solely on precursor masses for annotation generated many incorrect peak annotations. One frequently overlooked source of incorrect peak annotations is isomers produced alongside chemical reference standards, even when these were marketed and purchased as 'pure standards'. Other potential reasons for false peak annotations by MS1 peak picking encompass in-source fragmentations, artifacts from chemical noise and buffer chemicals, or cross-contamination between samples. To ensure the integrity and comprehensiveness of curated libraries, we first removed all MS1 candidates that did not yield information-rich MS/MS spectra, using spectral entropy S<0.5 as the threshold. This step removed 0.03-2.87% of all candidates (Table 1). Next, we developed a deduplication algorithm to identify the authentic spectra associated with both target compounds and their isomers, while excluding spectra originating from artifacts and other contaminant sources (Figure 2). As it is reasonable to assume that the most abundant peak ('major compound') corresponds to the pure reference chemicals, such peaks were assigned as the most probable source for the true MS/MS spectrum. Lower abundant chromatographic peaks within a 10 s range and within 5 mDa precursor mass error were labeled as isomeric impurities ('minor compounds'), if such peaks were not detectable in preceding or subsequent samples. Isomeric peaks with retention time differences > 10 s were retained if peak intensities exceeded 0.33 base peak intensity, and if MS/MS spectra exhibited an entropy similarity of > 0.75 to the major compound.

Table 1. Overview of retrieving and curating library entries using LibGen for EAD, HCD and UVPD libraries.

Libraries	EAD library			UVPD			HCD		
# compounds injected	1614			2007			2007		
Libraries overviews	# unique compounds	# MS[1] candidates	% compounds remaining	# unique compounds	# MS[1] candidates	% compounds remaining	# unique compounds	# MS[1] candidates	% compounds remaining
Precursor Matched	1423	5737	88.2%	1729	13009	86.1%	1729	13009	86.1%
Low entropy curated	1423	5735	88.2%	1727	12919	86.0%	1717	12635	85.6%
Deduplicated	1423	3808	88.2%	1727	3797	86.0%	1717	3750	85.6%
Curated library	1343	3524	83.2%	885	1223	44.1%	1281	2220	63.8%

For the three datasets, the algorithm reduced the number of potential candidate MS1 features between 34 - 71% (Table 1). This deduplication step is essential for the subsequent recalibration process. By incorporating major compounds together with their plausible isomeric analogs, the coverage of curated libraries is considerably expanded (Figure 2).

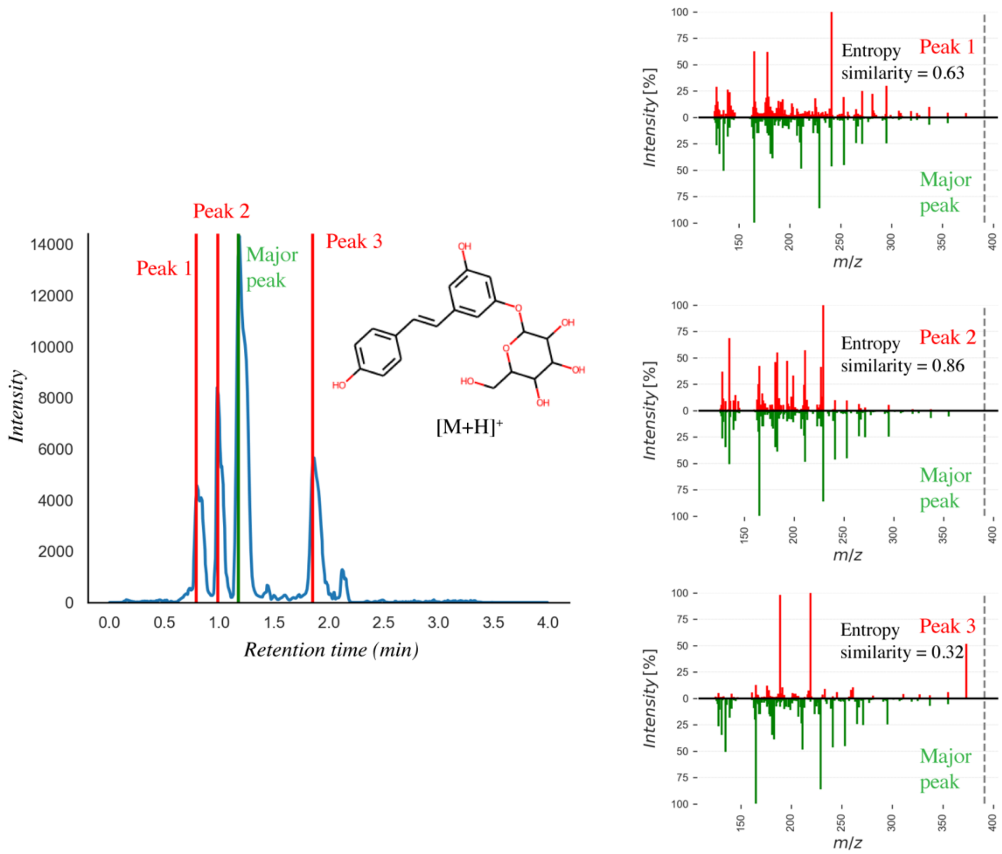

Figure 2. Deduplication of isomeric peaks in library curation. *Left*: extracted ion chromatogram m/z 391.139 for piceid as chemical reference standard, yielding four matching MS1 peaks. The major peak at 1.2 min is annotated as reference compound, while minor peaks at 0.7, 1.0 and 1.8 min represent isomers. *Right*: corresponding MS/MS spectra of minor peaks are matched by entropy similarity to the major peak. Isomers at entropy similarity > 0.75 are included into the final library.

All accurate mass instruments exhibit both systematic and stochastic mass errors, especially for low intensity compounds with low poor ion statistics[25,26,27]. To mitigate such technical errors during the curation of MS/MS libraries, LibGen recalibrates the fragment ion mass-to-charge (m/z) values for each data file from the complement of annotated MS^1 peaks in each mixture. Interestingly, a simple linear regression to model the relationship between MS1 error and MS1 intensity did not improve overall accuracy. In the mass ranges used for our natural product libraries, no relationship between m/z values and m/z errors was observed. Instead, LibGen employs a random forest model to predict the mass errors depending on the ion intensities, which is essentially a non-linear regression without specifying the kernel function. Incorporating retention time data did not enhance model performance. This mass recalibration resulted in a substantial improvement in mass accuracy across all three libraries (Figure 3).

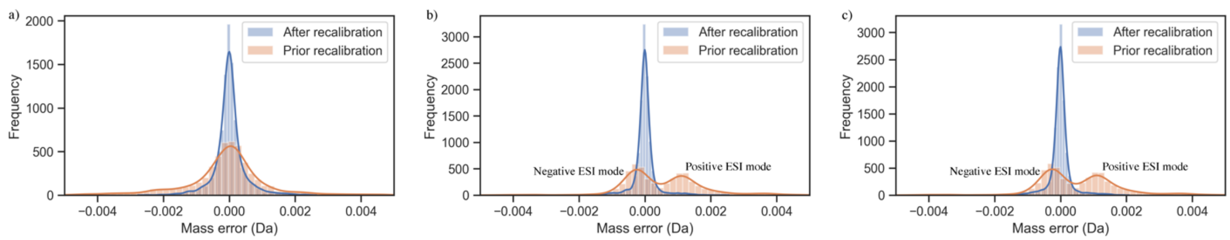

Figure 3. Kernel-density plots of MS^1 mass errors before and after mass recalibrations using random forest models of MS^1 precursor intensities: a) EAD dataset, b) UVPD dataset and c) HCD dataset.

This MS/MS mass recalibration is essential for the next step in the LibGen algorithm, denoising MS/MS spectra. Spectral denoising selectively eliminates false positive fragment ions in MS/MS spectra and ensures overall high quality of the library entries, especially for low abundant compounds. Here, LibGen employs a subformula-based spectral denoising algorithm[14, 28]. All fragment ions are validated by chemical plausibility, determining whether the exact mass of each fragment can be logically associated with a subformula loss from its parent molecular ion species. In addition, we expanded this logic to cover observations that in rare cases (<1%), the collision gas nitrogen may be added to formulas of fragments of aromatic compounds[29]. Similarly, oxygen atoms can be added to fragments of aromatic compounds during collision-induced dissociation[29]. When curating the natural product libraries presented here, we confirmed the rare occurrence of such fragments at <1%. Mass losses between molecular ion species and fragment ions were first searched against a database of 3.5 million chemical obtained from more than 26 chemical databases[30] that was increased to 10 million formulas by adjusting the number of hydrogens to cover radical ions (odd- and even-electron counterparts), in analogy to the BUDDY algorithm that was recently published to calculate elemental compositions for molecular ion species[30]. Only if fragments could not be rationalized against this formula database, the algorithm decomposes the molecular formula itself and comprehensively calculates all elemental subsets that could fit the mass loss while excluding implausible losses such as ones only account for pure carbon or pure nitrogen losses (except N_2)[31]. Such calculations are much more computationally expensive, and therefore only used after the direct matching. Hence, all fragments that cannot be rationalized by these two methods must logically come from another source. Such ions were removed from the experimental spectra. Consequently, for each library spectrum we calculated the residual explained intensity (EI%) as a relative measure of spectral quality by examining the percentage of valid MS/MS fragment intensity over the raw (uncurated) MS/MS fragment intensity:

$$Explained\ intensity\ (\%) = \frac{\sum_{p,valid} I_{p,valid}}{\sum_{p} I_{p}}$$

Where I_p is the intensity of a given peak.

To remove low-quality spectra from MS/MS libraries, LibGen applies both two criteria: (*1*) $S > 0.5$, to ensure that spectra with sparse fragmentation do not get added to the library, e.g. if only one or two fragment ions are generated. (*2*) Explained intensity EI > 90%, to ensure that the vast majority of fragments in a spectrum can be assigned to the parent molecular structure.

Only few spectra were removed by the low entropy criterion (*1*), removing between 0.1-3% of the total number of MS/MS spectra (Table 1). However, in comparison, a surprisingly high number of spectra showed a high number of fragment ions that could not be logically explained by elemental formula losses. For the EAD library, we found that 5.6% of the spectra were removed in the final denoising step (Table 1), giving an overall high quality of the spectra generated by this method. Surprisingly, the HCD library contained 40.8% spectra that had to be removed in the %EI denoising step (Table 1). Even more spectra were removed for the UVPD dataset, with 67.8% of all spectra giving less than 90% of explained fragment ion intensities (Table 1).

EAD spectra may be interpreted as being of overall higher quality because the EAD mechanism combines two powerful fragmentation pathways, a reaction between small, singly charged ions and electrons followed by classic collision-induced dissociation[32]. As a consequence, the population of precursor ions is more efficiently subjected to fragmentation than by CID alone, yielding relatively contributions of noise and artifact ions in MS/MS spectra. Correspondingly, when analyzing the CID MS/MS spectra, a high proportion of unfragmented precursor ions was observed, leading to an increase in the relative contributions of unexplainable MS/MS noise ions. Similarly, UVPD ionizes ions through the absorption of high-energy photons with a predetermined wavelength, selectively breaking covalent bonds and producing unique fragmentation patterns at low fragmentation efficiency, often leaving much of the precursor ion population unfragmented[12]. Specifically, for compounds lacking conjugate systems, the energy absorption process is significantly reduced, yielding lower fragmentation efficiency for these compounds that lead to MS/MS spectra with a high ratio of noise ions, which are subsequently removed by the denoising algorithm[12].

Validation of the denoising algorithm on high/low quality datasets

Next, we tested the effectiveness of the LibGen denoising algorithm by comparing 240 compounds in six highly-concentrated mixtures against 13-fold diluted mixtures using HCD fragmentation (Table 2, Supplement Method 2). As expected, only about half of the compounds in the 13-fold diluted dataset triggered MS/MS spectra acquisitions, yielding only about 1/3 of the MS/MS spectra compared to the positive control of all 214 compounds in the highly-concentrated dataset (Table 2). Here, we tested the hypothesis that spectra from diluted mixtures have a higher ratio of noise ions in MS/MS spectra, yielding a lower proportion of explained intensity. As comparison to LibGen's approach using chemical plausibility to differentiate between noisy and clean mass spectra, we also calculated normalized entropy as data-driven method[27]. In the highly-concentrated MS/MS dataset, LibGen labeled 77% of the spectra as clean spectra using an explained MS/MS abundance ratio of 90% as threshold (Table 2). For normalized entropy, we used a 0.8 cutoff as proposed previously, yielding an 89% rate of MS/MS spectra labeled as clean (Table 2).

Conversely, for the 13-fold diluted spectra, LibGen rejected 96% of the spectra, proving our hypothesis that noise ions in low-abundant MS/MS spectra have a significant contribution to overall MS/MS spectra and should therefore not be used in library building. In comparison, using normalized entropy only 50% of the highly diluted MS/MS spectra were rejected. Using normalized entropy therefore might lead to erroneous incorporation of poor-quality MS/MS spectra into libraries for compounds that inadvertently showed low ionization efficiency. As expected, MS/MS spectra with high normalized entropy values also generally showed lower explained intensity for both the highly-concentrated and the 13-fold diluted dataset (Figure 4A, 4B). However, a number of MS/MS spectra with very low explained intensity were still categorized as clean by normalized entropy. Such spectra contained many noise ions that could not be justified by their constituent elements, and the denoising algorithm accurately identified them as invalid (Figure 4C). Similarly, the LibGen denoising algorithm qualifies a range of spectra as good-quality although they have high normalized entropy (Supplement Data 1). This data exemplifies the utility of the LibGen denoising algorithm as essential to exclude low-quality MS/MS spectra while avoiding to excluding information-rich true positive MS/MS spectra.

Table 2. Performance of quality control by subformula denoising algorithm and normalized entropy on the concentrated and diluted datasets.

	# compounds	# spectra	# clean spectra by normalized entropy	# clean spectra by subformula denoising
Highly concentrated set	214	1049	935	807
13-fold diluted set	106	347	175	13

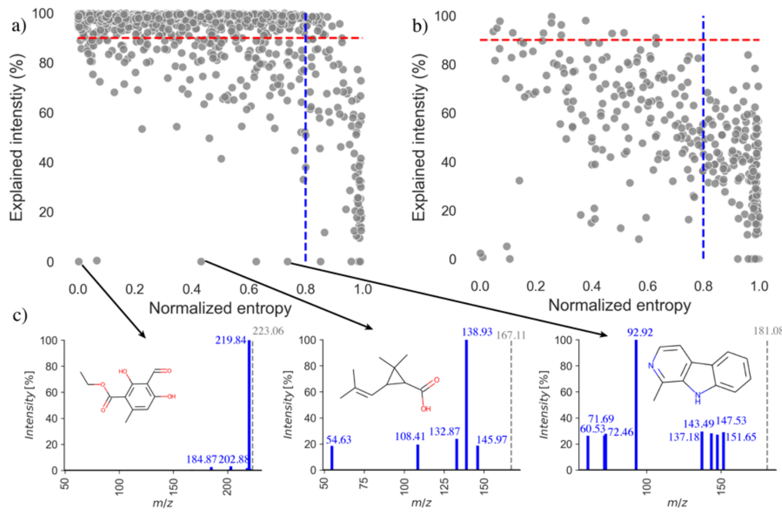

Figure 4. Benchmarking results for MS/MS quality assessments using a dataset of highly-concentrated compound mixture (a) and a 13-fold diluted dataset of the same compounds (b). MS/MS quality thresholds are used as 0.8 for normalized entropy (blue dashed lines) and 90% explained intensity (red dashed lines) for the LibGen subformula denoising algorithm. (c) Three example MS/MS spectra are given for the highly concentrated dataset with low normalized entropy but no chemically plausible ions (from left to right: haematommic acid $[M-H]^-$, ethyl ester, chrysanthemic acid $[M-H]^-$, and harmane $[M-H]^-$).

Overview and comparison of curated libraries

The three libraries generated here served slightly different purposes and were acquired separately. Therefore, the chemical diversity of the EAD library was different to the UVPD/HCD libraries, although a number of natural products were shared. Overall, 486 compounds were detected in all three libraries, while 368 were only tested and detected in the EAD library, while 50 compounds were exclusively found in the UVPD library and 95 compounds solely presented in the HCD library (Figure 5a). By utilizing the 2048-bit Morgan fingerprint and Tanimoto similarity index for chemometric analysis, all libraries displayed a high degree of structural diversity. More than 96% of all compounds in all three exhibited culmulative Tanimoto co-similarity indices < 0.2 (Supplement Data 2). Around 22% of all compounds were only detected as adducts other than $[M-H]^-$ or $[M+H]^+$, highlighting the necessity of considering all adduct types when building mass spectral libraries (Supplement Data 3). Importantly, the curated EAD and UVPD libraries are the first metabolomic libraries of their kind and may serve as test data to study fragmentation mechanisms. Therefore, all three libraries are open-access and can be publicly downloaded from MassBank.us.

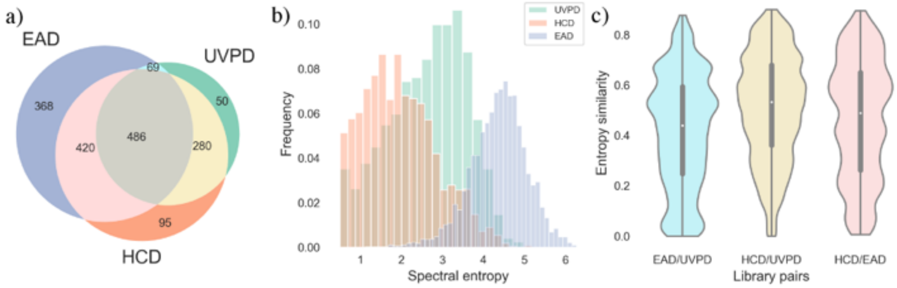

Figure 5. Key statistics of three curated libraries: (a) Compound overlap between the EAD, HCD and UVPD libraries. (b) Normalized distribution of the spectral entropies of the MS/MS libraries (c) MS/MS similarity of compound spectra across libraries.

The use of UVPD and EAD has not been explored comprehensively for annotating natural compounds, specifically in comparison to classic collision induced dissociation. We found that EAD spectra showed far richer fragmentation spectra than UVPD or HCD spectra with an average spectral entropy of 4.42, with 90% of all spectra found between spectral entropy 3.3-5.5 (Figure 5b). Similar to other small molecule libraries, the natural products used here showed low entropy in HCD fragmentation, with an average of spectral entropy 2.1 and a 90% range from 0.8-3.6. UVPD spectra were more information-rich than HCD spectra but showed clearly less fragmentation than EAD spectra (Figure 5b), with a mean spectral entropy of 2.8 and 90% quantile ranging from 1.1 to 4.1. Hence, both EAD and UVPD fragmentations are better suited than HCD to yield unique fragmentation patterns that can be used to identify natural products in untargeted metabolomics. Third, we found that MS/MS spectra of the same compounds showed overall little similarity across the three instrument types (Figure 5c). Here, we quantified MS/MS similarity in a pairwise manner using entropy similarity. Few compounds showed highly similar MS/MS spectra at entropy similarity > 0.75 between the different instrument types, while around half of the compounds showed at least moderate similarity between fragmentation types at entropy similarities between 0.6-0.75 (Figure 5c). Often, some fragment ions are shared between two fragmentation types, while even abundant ions may be absent from the comparator MS/MS fragment spectra (Figure 6). Hence, acquiring complementary and orthogonal fragmentation data through MS/MS analysis may significantly enhance confidence in compound identification in complex mixtures and add to our understanding of fragmentation.

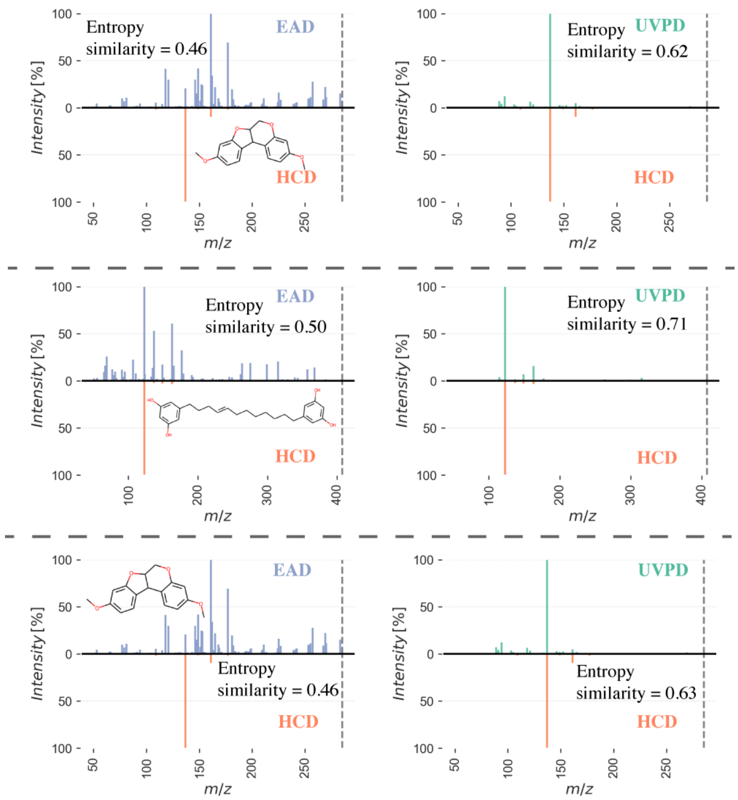

Figure 6. Head-to-tail comparisons of EAD/HCD mass spectra (left panels) and UVPD/HCD mass spectra (right panels) for three example compounds. *Top*: Homopterocarpin, $[M+H]^+$. Middle: 5-[(Z)-12-(3,5-dihydroxyphenyl)dodec-8-enyl]benzene-1,3-diol, $[M+NH_4]^+$. *Middle*: (1S,4aS,5R)-5-[2-(furan-3-yl)ethyl]-1,4a-dimethyl-6-methylidene-3,4,5,7,8,8a-hexahydro-2H-naphthalene-1-carboxylic acid, $[M+Na]^+$. *Bottom*: Homopterocarpin, $[M+H]^+$. Dashed lines indicate precursor m/z.

ESI positive and negative modes in UVPD

While EAD is currently only available for positive mode electrospray ionization, we were also interested to evaluate the difference between positive and negative ESI mode spectra under UVPD fragmentation. For both, raw and curated MS/MS spectra, we found that UVPD fragmentation yielded about four-fold more positive ESI spectra than negative ESI spectra (Supplement Data 3). Also, UVPD produces more fragments in positive mode (Figure 6). Collectively, spectra obtained in positive mode carry higher information content, with a mean spectral entropy of 2.39 compared to a mean spectral entropy of 1.48 in the negative mode.

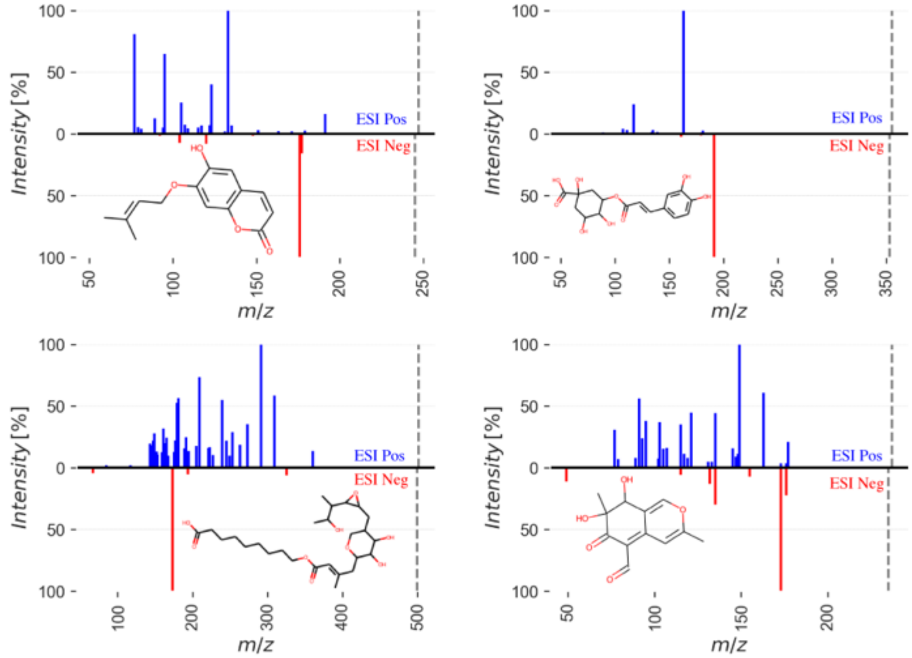

Figure 7. Comparisons of UVPD MS/MS spectra across ESI positive ([M+H]$^+$, blue) and negative ([M-H]$^-$, red) modes with selected compounds: a) Prenyletin, b) Chlorogenic acid, c) 2-amino-5-[2-[[2,3-dihydroxy-2-(1-hydroxyethyl)butanoyl]oxymethyl]-4-hydroxyanilino]-5-oxopentanoic acid and d) furanone. The dashed line indicates the precursor m/z.

Annotation using curated libraries

To validate the capability of libraries constructed by LibGen, compound annotation was performed on biological samples. Human GI tract samples[33] were analyzed by LC-EAD mass spectrometry spectrometry using a ZenoTOF 7600 mass spectrometer, while the NIST bilberry SRM 3291 dietary supplement reference material was analyzed by LC-HCD/UVPD mass spectrometry using a Orbitrap IQ-X instrument. The latest version of MS-DIAL 4.9.2[34] was used for data preprocessing, and the MSP file was imported into LibGen for compound annotation via an identity search using entropy similarity. All MS/MS spectra were subject to cleaning prior to library searching, and a similarity score of 0.75 was employed for unambiguous compound annotation with a minimized false discovery rate[27]. Annotations using the curated libraries are given in Supplement Data 4, showing select examples in Figure 8.

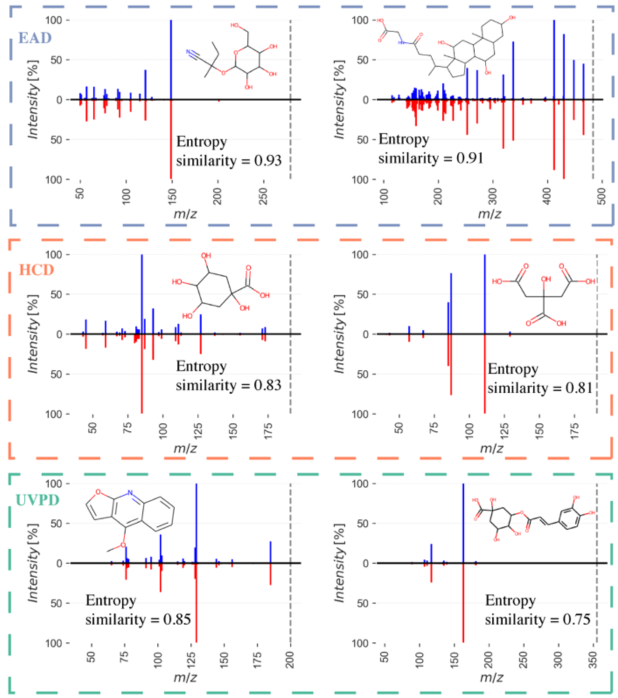

Figure 8. EAD spectra on human GI tract samples and UVPD/HCD spectra on bilberries and annotated with curated libraries. *Top*: EAD library, isomeric lotaustralin ($[M+NH_4]^+$, *left*) and glycocholic acid ($[M+NH_4]^+$, *right*). *Middle:* HCD library, quinic Acid ($[M-H]^-$, *left*) and citric acid ($[M-H]^-$, *right*). *Bottom:* UVPD library, dictamnine ($[M+H]^+$, *left*) and dictamnine ($[M+H]^+$, *right*) . The minimum spectral similarity for an annotation is 0.75.

4. Conclusion

Open-access spectral libraries play a crucial role in the advancement of metabolomics, particularly as the development of novel ionization techniques continues to gain momentum. In this study, we presented LibGen, a fully automated solution for curating MS/MS spectral libraries from mixes of reference standards with exceptional efficiency and reliability. Specifically, we demonstrated the superior performance of the subformula-based denoising algorithm compared to using normalized entropy for deciding which spectra to include in high quality libraries. Curated by LibGen, three specialized metabolite libraries comprising highly diverse chemical standards acquired by HCD-, EAD- and UVPD-fragmentation were curated. The EAD and UVPD natural product libraries are the first of their kind as publicly available MS/MS libraries. Being freely available on MassBank.us, users can employ such libraries for nutritional or gut microbiome research and related fields. Open access MS/MS libraries like shown here facilitate advancing informatics algorithms to better interpret fragmentation mechanisms and to identify unknown compounds, both of which require high-quality and high-confidence data. The application of the curated EAD and UVPD libraries is demonstrated for compound annotation on biological samples. The distinct fragmentation manner of EAD and UVPD shows potential opportunities for future researchers to use these tools as complementary evidence to classic collision induced dissociation spectra for compound identifications.

Data availability

All curated libraries are uploaded onto MassBank of North America database (Massbank.us) and can be freely downloaded. The molecular formula database is uploaded to the project repository on GitHub (https://github.com/FanzhouKong/Libgen_stable/tree/main/db). Raw spectra files can be found upon request.

Code availability

The source code of LibGen for curating spectral libraries with all associated modules can be found at https://github.com/FanzhouKong/Libgen_stable, along with demo data, Jupyter notebooks and documentation.

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Chapter 2: *Denoising Search* doubles the number of metabolite and exposome annotations in human plasma using an Orbitrap Astral mass spectrometer

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Abstract

Chemical exposures may impact human metabolism and contribute to the etiology of neurodegenerative disorders like Alzheimer's Disease (AD). Identifying these small metabolites involves matching experimental spectra to reference spectra in databases. However, environmental chemicals or physiologically active metabolites are usually present at low concentrations in human specimens. The presence of noise ions can significantly degrade spectral quality, leading to false negatives and reduced identification rates. In response to this challenge, the *Spectral Denoising* algorithm removes both chemical and electronic noise. *Spectral Denoising* outperformed alternative methods in benchmarking studies on 240 tested metabolites. It improved high confident compound identifications at an average 35-fold lower concentrations than previously achievable. *Spectral Denoising* proved highly robust against varying levels of both chemical and electronic noise even with >150-fold higher intensity of noise ions than true fragment ions. For human plasma samples of AD patients that were analyzed on the Orbitrap Astral mass

spectrometer, *Denoising Search* detected 2.3-fold more annotated compounds compared to the Exploris 240 Orbitrap instrument, including drug metabolites, household and industrial chemicals, and pesticides. This combination of advanced instrumentation with a superior denoising algorithm opens the door for precision medicine in exposome research.

Introduction

Human diseases are influenced by both genetic predispositions and environmental factors (GxE). Environmental impacts, including diet, lifestyle, and biological and chemical exposures, account for over 70% of disease incidence[1]. However, chemical exposures in human samples are usually low abundant at trace levels, similar to levels of physiologically active metabolites such as oxylipins, endocannabinoids or modified bile acids[2, 3]. Nontargeted exposome research as well as metabolomics and lipidomics methods rely on liquid chromatography coupled with high resolution mass spectrometry (LC-MS/MS)[4]. These small molecules are identified by matching experimental spectra against established repositories like the NIST23 library, MassBank of North America (MassBank.us), or GNPS/MassIVE[5, 6]. While the quality of the experimental spectra is critical for accurate MS/MS matching, mass spectra are often compromised by both electronic noise and chemical noise [7-9]. This problem is particularly pronounced in metabolomics and exposome studies, where the prevalence of low-abundance compounds can greatly exceed that of high-abundance compounds[2]. Electronic noise originates from the inherent characteristics of the electrical system, the discrete nature of ion signals, or the process of Fourier transformation[7, 10]. Chemical noise is derived from components in the sample that confound the signal generated by the metabolites and exposome compounds of interest[9]. Chemical noise in LC-MS/MS is defined as isobaric interferences that emerge from the testing materials themselves, or from laboratory consumables, solvents, cross-contaminations, buffers or carryovers[9]. The amalgamation of true and contaminant fragment ions produces chimeric spectra. Chimeric spectra drastically affect spectral matching scores and result in false negatives peak annotations, contributing to the ‘dark matter of metabolomics’[11, 12]. In proteomics, methods for noise removal resort to intensity modeling[13, 14] and matching to in-silico spectra[15]. In nontargeted small-molecule studies, methods were developed that required specific experimental conditions to monitor the precursor-fragment ratios[16], or leveraging database assistance[17, 18]. Overall, these methods provided only modest enhancements and low throughput. In addition, public datasets[19] frequently lack the experimental settings and database metadata required for these methods. Consequently, existing denoising methods are unsuitable for large scale, standardized de-noising in metabolomics. Typical metabolomics data processing software denoises spectra by simply discarding ions below 0.5-1% of the base-peak height[20-22]. Surprisingly, the chemistry information revealed by the fragment peaks are not considered when determining if a given fragment is true ion or noise. We here show that integrating intensity modeling with assessing chemical plausibility greatly enhances the effectiveness of noise ion removal, termed *Spectral Denoising*. *Spectral Denoising* first eliminates electronic noise by stratifying fragment ion intensities, followed by filtering the remaining fragments based on their chemical plausibility as true fragments of the molecular formula of the precursor ion. Utilizing a 13-stage series dilution dataset of 240 small molecules generated an experimental benchmarking dataset with varying levels of spectral quality. This dataset was used to rigorously benchmark the robustness of *Spectral Denoising* against other methods, including by virtually adding different levels of contaminating chemical and electronic noise ions. False discovery rates (FDR) were thoroughly tested against 1,267 experimental spectra that were annotated by NIST23, MassBank.us and GNPS libraries. By integrating *Spectral Denoising* into the spectral matching process ('*Denoising Search*'), we evaluated its performance using human plasma samples from AD patients acquired with advanced Asymmetric Track Lossless (Astral) mass spectrometry and classic Orbitrap instruments. The number of annotated compounds increased more than 2-fold with *Denoising Search* compared to classic compound annotation pipelines. By combining Astral mass spectrometry and *Denoising Search,* low-abundance exposome compounds can be detected in human plasma that have not been reported before in the literature. Hence, *Denoising Search* may herald a new era in the identification of key biomarkers, more confident compound annotations and better interpretability of datasets that are critically needed for biomedical research like neurodegenerative diseases.

Results

Figure 1 gives the schema of how the *Spectral Denoising* algorithm removes ions recorded in collision-induced MS/MS spectra that do not represent genuine fragments of the precursor ion. The first step removes electronic noise that commonly appears as a multitude of (low-abundant) ions with very similar intensities, also termed 'grass noise'[23]. Chemical noise is harder to recognize because it is generated by co-isolating and fragmenting non-target precursor ions in low-resolution quadrupole mass filters that precede the collision-induced dissociation even in high-resolution mass spectrometers[24]. In practice, the isolation window used in most metabolomics studies ranges from 1 to 5 Da, increasing the likelihood of inclusion of contaminant ions due to isobaric interference[25]. These chemical noise ions can vary widely in intensity, making them difficult to distinguish from true fragment ions by manual inspection alone[9]. Hence, our schema to remove chemical noise ions is based on the unique property of true fragment ions to produce a chemically plausible neutral loss (or radical losses), calculated from the accurate mass of the target precursor ion[26] (Figure 1).

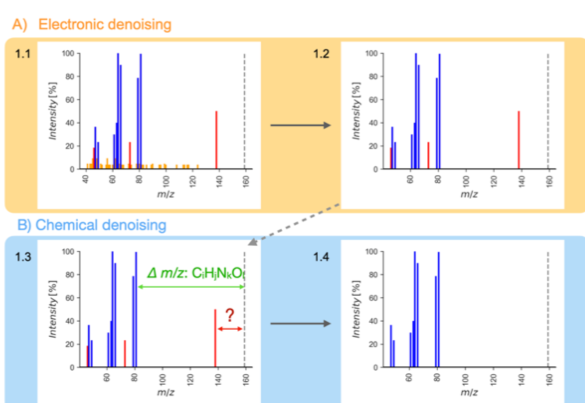

Figure 1. Flowchart for *Spectral Denoising*.
(**a**) Removing electronic noise by recognizing repeated fragment ions with identical intensities.
(**b**) Removing chemical noise by identifying remaining fragment ions that do not fit possible elemental subformulas from the precursor ion mass. The dotted line indicates the precursor ion *m/z*.

To test this concept, we first empirically probed all 230,000 MS/MS spectra of the NIST23 library that was generated by the U.S. National Institute of Standards and Technology through a rigorous validation and fragment verification process to guarantee a high fidelity of data quality. For more than 99.5% of all ions per spectrum, fewer than four ions were found within relative intensities of ±0.1% (Extended Figure 1). This data served as valid threshold to automatically identify electronic noise ions in experimental MS/MS spectra and remove these ions, independent of the relative intensity (Figure 1A).

Electronic denoising differs in two key aspects from simply discarding ions below a pre-defined threshold. First, ions are not discarded simply based on relative intensity levels. In this way, electronic denoising retains low-abundant ions that may represent true fragment ions of the precursor molecule. This step is important because many small molecules do not produce fragment-rich spectra, unlike peptides, covered in the new concept of spectral entropy[21, 27], denoted as *S*, which is a relative measure of the richness of information. The denoised MS/MS example spectrum in 1.4 (Figure 1B) would calculate *S*= 1.9, compared to a more disordered contaminated spectrum 1.1 with *S*=3.3 before the denoising process. Second, electronic noise becomes relatively more prominent for MS/MS spectra that originated from very low-intensity precursor ions. Therefore, a simple cut-off threshold at 1% base peak intensity does not suffice for metabolomics or exposome nontargeted studies that aim at low abundant molecules[20, 22].

Subsequently, chemical denoising identifies and removes chemical noise ions by calculating whether the exact mass of each fragment can logically be associated with a subformula loss from the parent molecular ion species (Figure 1B). The chemical plausibility of relative loss subformulas was validated using the Seven Golden Rules algorithm to discard chemically impossible losses (e.g., CH_{12})[28] while ignoring the LEWIS and SENIOR checks that are designed for intact molecules. In collision-induced dissociation, a fragment ion can result from multiple relative losses from the precursor ion, potentially violating the SENIOR rule[29].

Instead, chemical denoising expands the logic of our subformula-loss calculations of chemical noise ions. For example, radical fragment ions are formed in about 10% of small molecule MS/MS spectra as metastable state in mass spectrometry, even for even-electron precursors[30]. Enforcing the LEWIS rule would lead to the removal of these valid radical fragment ions[30]. Moreover, about 1% of MS/MS spectra were reported in which the collision gas nitrogen formed bonds with substituted aromatic compounds within the collision cell, with subsequent background water substitution[31, 32]. We confirmed the occurrence of such fragmentations in the NIST23 spectral library and validated these experimentally in our laboratory. Hence, *Spectral Denoising* accounts for possible [$M+N_2+H_2O$] molecule reactions when calculating the relative loss of fragment ions for substituted aromatic compounds. The sequential combination of electronic and chemical denoising was validated on 10,000 NIST23 spectra and compared MS/MS similarities of the denoised against the original library spectra (Extended Figure 2). Entropy similarity is scaled in the same manner as classic dot-score similarities, from 0-1 with 1 marking perfect matches and 0 giving no similarity at all. If no ions were removed by the denoising algorithm, MS/MS similarities would remain identical, leading to perfect matching scores of 1. In all calculations, the remaining abundance of the precursor ion intensities is ignored to focus entirely on genuine fragment spectra, and because all identity-search algorithms already exclude compounds that do not match specific accurate mass windows of the precursor mass (typically at 5 ppm). The average similarity of the selected denoised NIST23 spectra was >0.99, proving that the denoising method did not accidentally remove true ions, and that the NIST23 spectra have very high quality (Extended Figure 2).

Denoising experimental MS/MS spectra of 240 metabolites diluted from 500-0.02 pmol injections

To evaluate the effectiveness of our denoising strategy, we analyzed 240 metabolites in both positive and negative electrospray ionization (ESI) modes in a 13-step serial dilution from 500 pmol to 0.02 pmol injected onto the column. MS/MS spectra of the most concentrated 500 pmol injections represented optimal spectral quality, while the more diluted ones were expected to gradually deteriorate in spectra quality due to an increased contribution of contaminating noise ions. A total of 28 compounds were excluded due to insufficient fragmentation with less than two fragment ions (spectral entropy <0.5). The remaining dataset of MS/MS spectra that were used for MS/MS similarity calculations included a total of 11,823 spectra, encompassing 6,885 in the positive mode and 4,938 in the negative ESI mode (Extended Figure 3).

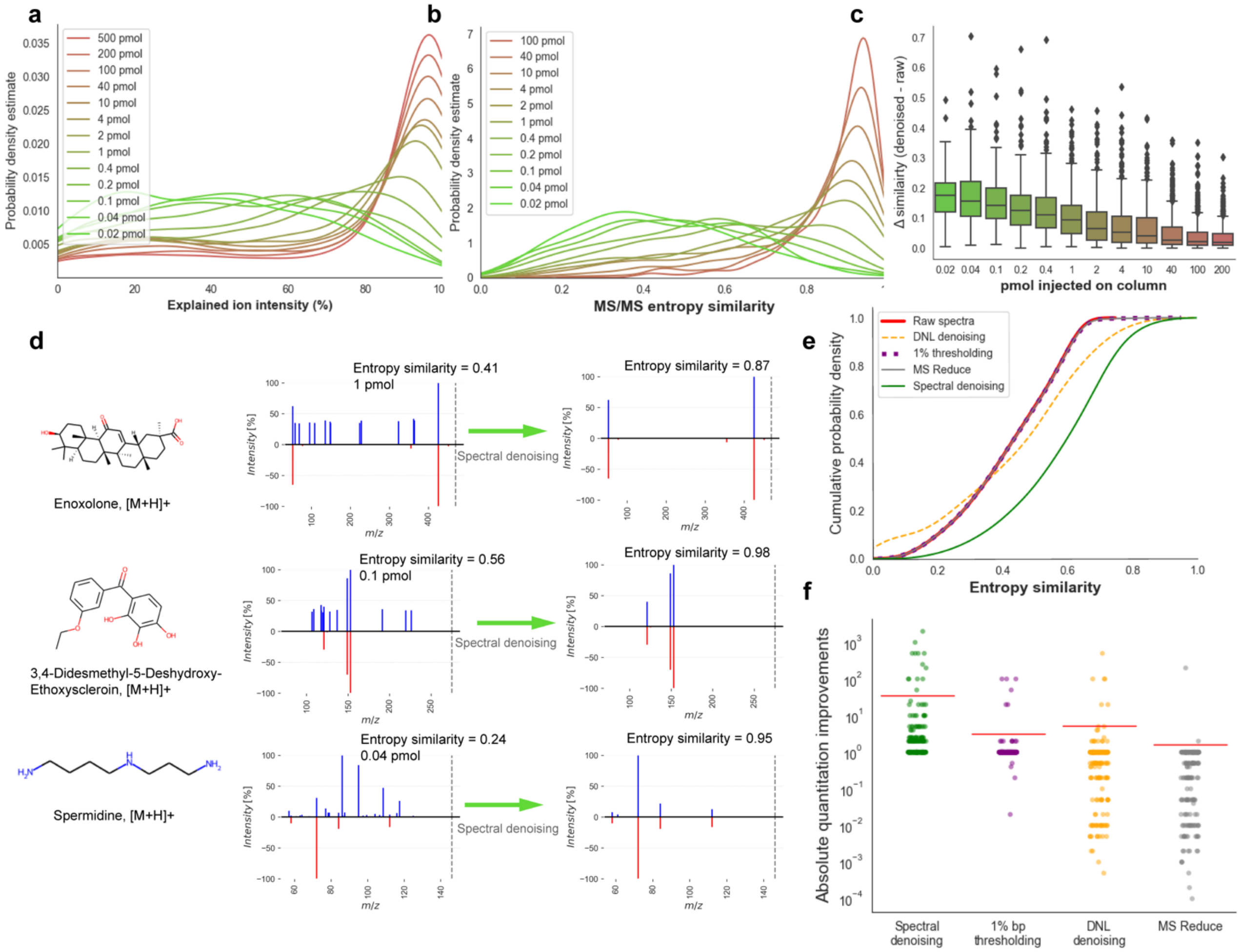

Figure 2. Developing, validating and benchmarking the *Spectral Denoising* algorithm. (**a**) Probability density distribution (explained denoised/raw intensities) of all MS/MS spectra from 240 injected standards between 0.02-500 pmol. **(b)** Probability density distribution of the entropy similarities before *Spectral Denoising* of all MS/MS spectra from 240 injected standards between 0.02-200 pmol, using the 500 pmol spectra as reference. **(c)** Improvement in spectral entropy similarities after *Spectral Denoising*. **(d)** Examples of head-to-tail plots of MS/MS spectra before and after *Spectral Denoising* for compounds injected at low quantities. **(e)** Cumulative distribution of MS/MS entropy similarities before (‘raw’) and after applying three benchmarking methods against the *Spectral Denoising* algorithm. Only raw spectra with MS/MS entropy similarities <0.75, a typical threshold for automatic metabolite annotations. **(f)** Strip plot to visualize absolute improvements in MS/MS similarities across three benchmarking methods and the *Spectral Denoising* algorithm.

First, the ability of the denoising algorithm was evaluated to discern noise ions from genuine fragment ions. To this end, the total explained ion intensity was enumerated as metric to quantify the proportion of true ion intensities in each spectrum (Figure 2a). The probability density of explained ion intensities shifted markedly from high to low amounts of injected compounds, highlighting the effectiveness of the denoising algorithm to identify noise ions. The probability distributions of the calculated entropy similarities (Figure 2b) showed and even more pronounced decrease in median MS/MS similarities with lowered injected amounts, from >0.92 entropy similarity at 200 pmol injected to <0.41 median similarity at the lowest injection quantities. Remarkably, at 1 pmol injections (about 0.3 ng injected onto the column, at the median molecular mass of the chemicals included test mixture), more than 50% of all MS/MS spectra already failed to match the reference spectra at entropy similarity > 0.75, a cut-off that is often used in metabolite annotations in metabolomics (Figure 2b). More importantly, the *Spectral Denoising* algorithm effectively removed chemical and electronic noise ions for all test compounds (Figure 2c). As expected, the largest improvement for MS/MS similarity calculations were observed for very low injected amounts with a median gain of 0.18 entropy similarity scores. At 1 pmol injections, a median improvement of 0.1 MS/MS entropy similarity gain was noted, and even for 200 pmol injections, 25% of the compounds already showed an improvement of entropy similarities of 0.05 (Figure 2c). Example spectra are depicted in Figure 2d with the precursor ions given as dotted lines to indicate that residual precursor mass intensities were ignored in MS/MS similarity calculations. For 3,4-didesmethyl-5-deshydroxy-ethoxyscleroin and enoxolone, raw spectra MS/MS of low abundant injections were notably marred by substantial electronic noise with up to 30% relative base peak intensity, vastly exceeding the typical 1% base peak ratio often used as a threshold (Figure 2d). For spermidine injected at 0.04 pmol, the initial raw spectrum displayed a diverse set of ion intensities without obvious signs of electronic noise. Surprisingly, the three most abundant ions m/z 86.004, m/z 95.009, and m/z 108.494 were identified as chemical noise, compromising the entropy similarity to a level of 0.24 score. The denoised spermidine spectrum showed a perfect match with an entropy similarity of 0.95 (Figure 2d).

Next, the efficiency of *Spectral Denoising* was evaluated by benchmarking its performance against three established MS/MS denoising techniques: the classic 1% base peak height thresholding method[20, 22], dynamic noise level (DNL) denoising[14], and MS Reduce denoising[13]. The benchmarking dataset comprised all 2,677 spectra that had raw MS/MS similarities <0.75 score, meaning they might not be annotated by high confidence identification schemas (Figure 2e). Similar to *Spectral Denoising*, the three methods selected for benchmarking purpose are standalone tools with a generalized scope of applicability, as they do not necessitate complementary metadata or additional experimental setups. While we initially hypothesized that all denoising methods might enhance spectral matching, neither of the three benchmarked algorithms showed any marked improvement. DNL denoising yielded only a slight improvement in entropy similarity for a limited subset of spectra, with a modest increment of 0.01 in spectral similarity. The MS Reduce approach, even at the highest quantization level of 11[13], failed to enhance spectral similarity effectively. Surprisingly, even the classic 1% base peak thresholding method showed negligible impact on spectral similarity matching, indicating that while simple and computationally inexpensive, this method is inadequate for noise ion removal in low-abundance compound spectra. In contrast, our *Spectral Denoising* method showed significant gains for MS/MS spectral matching with a median entropy similarity increase of 0.17, lifting more than 1,500 spectra to MS/MS similarity >0.75 and thereby boosting the compound annotation rates by 30% (Figure 2e).

A second benchmarking test quantified at how much lower injected quantities compounds could be annotated at entropy similarity >0.75 by applying the *Spectral Denoising* method. For all 181 compounds that were detected in at least more than one dilution stage (Supplement 1), the fold-change was calculated between the lowest injected amount that reached >0.75 entropy similarity for the raw MS/MS spectra, compared to the lowest injected amount after *Spectral Denoising* (Figure 2f). On average, our *Spectral Denoising* method required 35-fold lower molar quantities injected onto the column (Figure 2f, Supplement 1), while not a single compound failed to be annotated after *Spectral Denoising* (no false negatives). In contrast, all other denoising techniques showed minimal improvements for annotations injected at lower absolute quantities. Specifically, the 1% base peak thresholding method demonstrated no enhancement for 163 compounds. Similarly, DNL denoising and MS Reduce only showed improvements for so-few compounds. More concerningly, both DNL denoising and MS Reduce did not only fail to improve quantity thresholds for successful MS/MS annotations but instead detrimentally affected spectral matching. For MS

Reduce, 108 compounds became unannotated (false negative) even at the same concentration level after processing the raw MS/MS spectra. For DNL denoising, this number of false negatives was 86 compounds. This indicates that both methods inadvertently removed true fragment ions, thereby shifting entropy similarity distributions to lower values across all spectra (Extended Figure 4). This disparity in performance across denoising techniques may originate from their foundational assumptions, as both DNL and MS-reduce were introduced on proteomics data that are usually fragment-rich, unlike in metabolomics where collision-induced fragments spectra are usually sparse. Thus, the underlying assumptions of the DNL- and MS-reduce intensity modeling-based denoising methods are no longer valid when applied to metabolomics data, highlighting the necessity for specialized approaches in this field.

Applying Spectral Denoising against artificial noise ions

Contamination by noise ions in experimental spectra from biological samples is more challenging than the sets of chemical standards shown above. A larger diversity of noise origins, e.g. from the chemosphere of the exposome, requires better mimicking large-scale contribution of different types of noise. To thoroughly assess the robustness of the tested denoising algorithms, the 0.01-200 pmol dilution series experimental spectra were used to create artificial chimeric spectra by adding simulated levels of chemical and electronic noise. Both types of noise ions were introduced using established noise models[8], albeit with different parameter sets to accurately reflect their characteristics. Chemical noise, characterized by real chemical formulas, typically appears with high intensity but low ion counts. The mass-to-charge ratios of chemical noise ions were sampled from a database of 3.5 million authentic chemical formulas[33], with their relative intensities determined using the noise model with a mean intensity of 50% of the base peak height. We defined three contamination levels for chemical noise, with the total noise ion counts to raw ion counts in a ratio of 1:10 (low), 2:10 (medium), and 5:10 (high). Electronic noise typically manifests as low-intensity but high-quantity 'grass noise'[23]. Thus, the mass-to-charge ratios of electronic noise ions were randomly sampled, with their relative intensities determined by the same noise model mean ion intensity of 5% base peak height. Similarly, we also designed three levels of electronic noise contamination: low (2:1), medium (10:1), and high (100:1), yielding nine tests of combined noise levels.

The resulting chimeric spectra were matched against the 500 pmol benchmark spectra of the authentic standards, and denoised with *Spectral Denoising*, MS Reduce, DNL denoising and 1% bp thresholding (Figure 3). The distribution frequency plot of all combined raw spectra of the compound dilution series yielded a median entropy similarity of 0.8, with an average of 0.71 and a mode at 0.95 (Extended Figure 4). Adding virtual noise to render chimeric spectra drastically reduced the spectral quality in all nine test scenarios, even for the lowest level of chemical and electronic noise (Figure 3), to a mode of 0.5 spectral entropy. When considering the mode points in the frequency distributions of the raw spectra, electronic noise worsened spectral entropy scores more dramatically than chemical noise additions, even at low levels of electronic noise. For all nine test cases, our denoising method restored the frequency distributions of the contaminated spectra above the levels of the original spectra, with the median spectral entropy similarity ranging from 0.71 (high electronic, high chemical noise) to 0.87 (low electronic, low chemical noise) (Figure 3). Importantly, the benchmarking test clearly demonstrated that none of the other algorithms came close to the performance of our *Spectral Denoising* method (Figure 3), with the best frequency modes located at spectral entropy 0.6 for the MS Reduce method for the low electronic, low chemical noise test scenario.

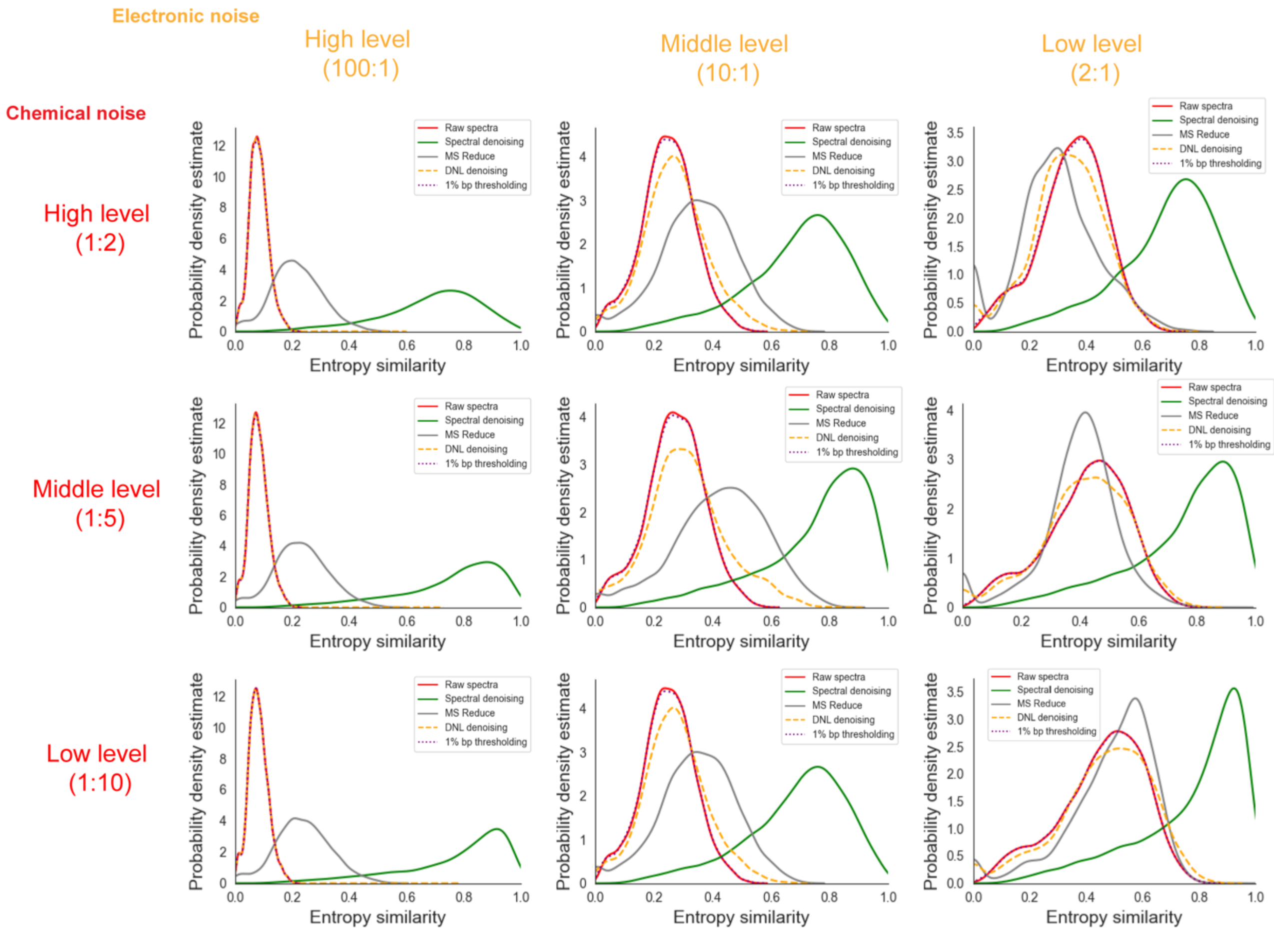

Figure 3. Probability distributions of MS/MS entropy similarities before ('raw') and after applying three benchmarking methods against the *Spectral Denoising* algorithm, under varying levels of artificially added chemical and electronic noises. <u>Chemical noise</u>: *Level 1:10* added at least one noise ion for every 10 experimental ions; *Level 1:5* added at least one noise ion for every 5 experimental ions; *Level 1:2* added at least one noise ion for every 2 experimental ions. <u>Electronic noise</u>: *Level 2:1* added two noise ions per experimental ion; *Level 10:1* added ten noise ions per experimental ion; *Level 100:1* added one hundred noise ions per experimental ion.

Overall, neither the 1% bp thresholding nor the DNL denoising approaches yielded any substantial improvement across any level of chemical noise contamination (Figure 3). Last, we investigated if the improvement of MS/MS similarities by *Spectral Denoising* depended on the entropy of the 500 pmol reference spectra themselves. Spectra were categorized into five groups, from 0-1 entropy (low number of fragment ions) to 4-5 entropy levels (high number of fragment ions with varying intensities). We suspected that most experimental spectra from biological samples would only be subjected to minor to moderate contamination and therefore only used mid-level and low-level combinations of virtually added noise ions.

As expected, reference raw spectra that started with lower entropy (0-1) benefitted the most from *Spectral Denoising*, as such spectra also see the most dramatic decline in MS/MS spectral similarities when noise ions are added (Figure 4). Conversely, reference raw spectra with high spectral entropy (S >4) better tolerate the addition of noise ions, and hence benefit a little less from *Spectral Denoising* (Figure 4). Yet, mass spectra from any starting entropy levels showed clear improvements in MS/MS similarity from *Spectral Denoising* when artificial noise was added, with a frequency distribution mode of 0.42 similarity improvements for S=4-5 spectra and middle levels of added noise, and 0.2 similarity improvements at low levels of added noise (Figure 4). The results for the remaining seven sets of entropy similarity improvements are given in Extended Figure 5. Overall, these sets of benchmarking and noise-addition experiments clearly demonstrates that our *Spectral Denoising* method outperforms all other techniques and is extremely robust across varying levels of chemical and electronic noise contamination.

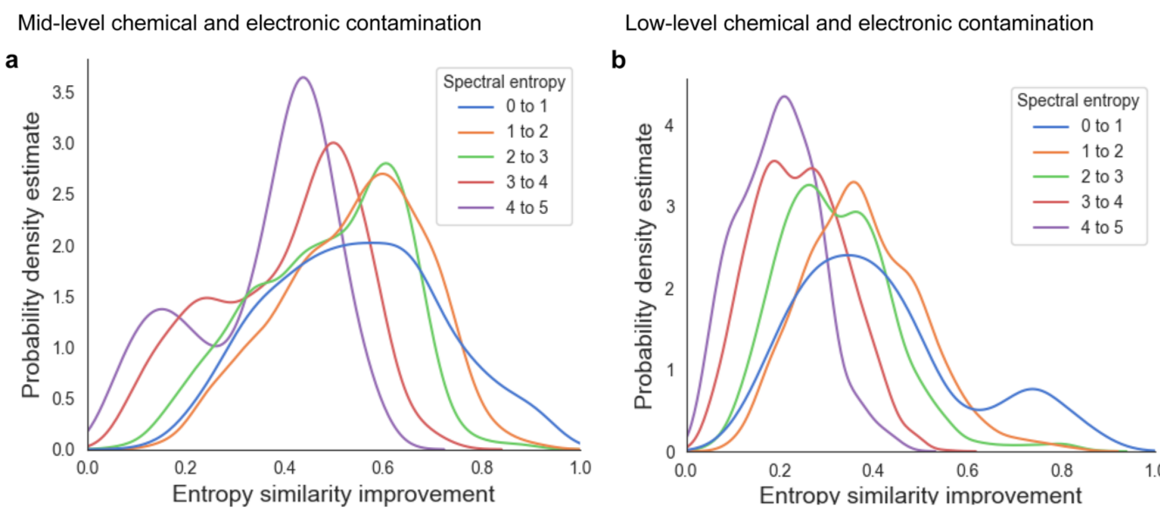

Figure 4. Density distributions for MS/MS similarity improvements after *Spectral Denoising* for all MS/MS spectra from 240 injected standards between 0.02-200 pmol, using the 500 pmol spectra as reference. Chemical standards were grouped into five sets with different starting spectral entropies (blue to purple). (**a**) MS/MS similarity improvements for spectra to which contamination ions were artificially added at '*mid-levels*', Level 10:1 electronic noise, and Level 1:5 chemical noise. **b** MS/MS similarity improvements for spectra to which contamination ions were artificially added at '*low levels*', Level 2:1 electronic noise and Level 1:10 chemical noise.

Development and applying Denoising Search on HILIC-MS/MS data from the plasma of AD patients

In the prior experiments, all test spectra had a priori knowledge of the molecular formula information. Although today's algorithms are capable of annotating molecular formulas from MS/MS spectra of reference compounds with >95% confidence[33, 34], these tests have never been conducted on low abundant or noisy spectra. To enhance the applicability of our denoising method, *Spectral Denoising* was integrated into the spectra searching process, now termed *'Denoising Search*.'

Denoising Search starts by denoising the experimental spectra using all molecular formulas that fall within the predefined precursor mass accuracy, i.e. not assuming a single starting formula. As it is a spectral identity search algorithm, it depends on the formula space that is being searched. In combination, MassBank.us, GNPS and NIST23 contain 2,028,556 experimental spectra, corresponding to 435,698 compounds and 37,493 formulas. When restricting the search space in this way, spectral matching scores for the *Denoising Search* were calculated based on all denoised

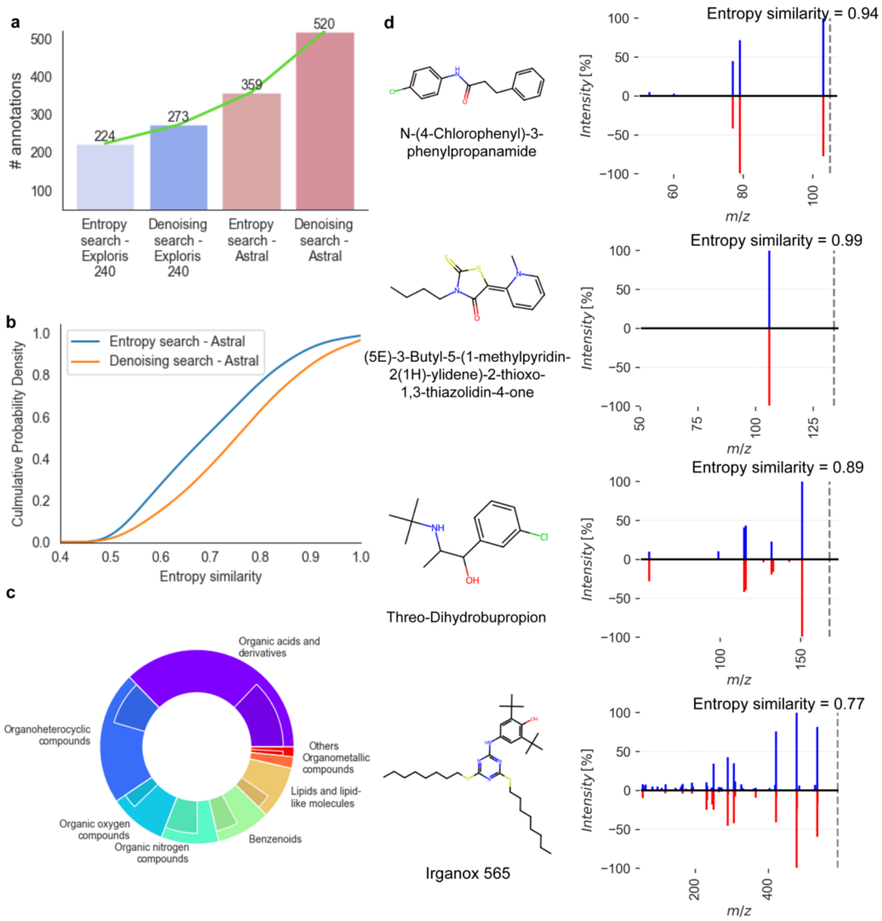

Figure 5. *Denoising Search* results for positive ESI mode HILIC-MS/MS data acquired on an Exploris 240 Orbitrap instrument and the Astral mass spectrometer, using 20 plasma samples of Alzheimer's disease patients. **(a)** Improvement of metabolite annotations at MS/MS similarity >0.75 before and after *Denoising Search* using MassBank.us, GNPS and NIST23 libraries. **(b)** Cumulative probability density before and after *Denoising Search* for Astral mass spectrometry spectra. **(c)** Proportions of metabolite annotations after *Denoising Search* at MS/MS similarity >0.75 for different chemical superclasses using the Exploris 240 Orbitrap (inner ring) or the Astral mass spectrometer (outer ring). **(d)** Head-to-tail plots for four selected compounds annotated uniquely on Astral data after *Denoising Search* (blue) against library spectra (red).

spectra that fit the formula criteria within the mass accuracy of the instrument. For practical reasons, a 10 mDa mass accuracy threshold was used although Orbitrap instruments are known to yield much better exact masses (i.e., sub-ppm with internal calibration). However, for low abundant ions, mass accuracy levels suffer in concordance with compromised ion statistics. Essentially, *Denoising Search* functions similarly to a Bayesian probability approach, evaluating how likely it is to observe the query spectra with all potential chemical and electronic noise removed, given a specific target compound. The rationale is that if the correct molecular information is used to denoise the spectra, noise ions will be accurately identified and removed, thereby improving spectral matching scores. Conversely, if the molecular formula information is incorrect, the fragment patterns will be vastly different, and the removal of true ions would lower the entropy similarity scores, still resulting in true negative annotations. This rationale was validated by testing the false discovery rate (FDR) of *Denoising Search* against simple entropy similarity identity search on an in-house validation dataset with extensive manual curation (Extended Figure 6). At an entropy similarity level of 0.75, the two methods showed <1% differences in terms of the FDR rate, indicating that *Denoising Search* did not introduce unwanted bias or false positives (Extended Figure 6).

To evaluate the performance of *Denoising Search* on experimental spectra of human patient samples, a small pilot plasma study was used comparing two high-resolution, accurate mass instruments: the classic Orbitrap Exploris 240 mass spectrometer and a new instrument introduced in 2023, the Orbitrap Asymmetric Track Lossless (Astral) mass spectrometer. The Astral mass analyzer acquired 17 times more spectra in each scan cycle compared to the Exploris 240 instrument, resulting in 5-times more MS1 m/z-retention time features that had corresponding MS/MS spectra. In effect, the Astral instrument provided a top-35 data dependent analysis MS/MS survey, surpassing the Exploris 240 instrument that only used a top-2 DDA mode. While the Orbitrap Astral instrument had previously shown its superior capabilities in proteomics studies, this comparison demonstrates its advantages for metabolomic tests. Overall, the raw spectra from the Astral analyzer achieved 60% more annotations than those from the Exploris 240 mass spectrometer when matching spectra against the NIST23, MassBank.us and GNPS libraries (Figure 5a).

For the Exploris 240 instrument, *Denoising Search* facilitated an additional 22% increase in annotations, while Denoising Search yielded a 45% increase in annotated compounds over the raw spectra for the Astral mass analyzer (Figure 5a). Hence, compared to the raw Exploris MS/MS spectra, the *Denoising Search* on Astral data led to 2.3-fold more annotations, including many exposome compounds that were not found on the Exploris Orbitrap instrument. Notably, using *Denoising Search*, a significant increase of 0.11 median MS/MS entropy similarity was achieved for Astral spectra that showed raw entropy similarity ≥ 0.4 (Figure 5b). A closer examination of the seven main ClassyFire compound superclasses revealed a notable increase in the number of annotations across all superclasses on denoised Astral spectra compared to those found with Exploris (Figure 5c). Superclasses such as organic acids and derivatives fully leveraged the capabilities of the Astral, resulting in a 2.2-fold increase in annotated compounds, while organoheterocyclic compounds saw an 89% increase in annotated compounds. A significant increase in the number of MS/MS spectra was acquired by Astral Orbitrap mass spectrometry, thanks to its high sensitivity and its unprecedented acquisition speed (up to 200 Hz in DIA mode, 160 Hz in DDA mode). By combining our Denoising Search with Astral mass spectrometry, several compounds were identified that were previously underexplored in human blood (Figure 5d). Beyond drugs like threo-dihydrobupropion and N-(4-chlorophenyl)-3-phenylpropanamide, Irganox 565, a hindered phenol antioxidant, was reported in human blood for the first time, despite its prior detection in environmental dust samples[35]. This pilot study demonstrates that the advancement of mass analyzers allows for the acquisition of more spectra, and the *Denoising Search* is crucial for fully taking advantage of these extra spectra triggered by precursors across a wide range of magnitudes.

Discussion

The impact of noise ions in spectra of low-abundance compounds is well-recognized in metabolomics and exposome research. These ions complicate chemical annotations, contributing significantly to the accumulation of 'dark matter' in small-molecule research. We here employed a strategy combining intensity modeling and subformula assignments to effectively eliminate noise ions while preserving essential true fragment ions, even at low relative intensities. Using a 13-stage dilution of MS/MS spectra of genuine chemical reference standards as a ground truth dataset, demonstrated a superior ability for the *Spectral Denoising* algorithm to identify and remove noise ions. Noise removal notably enhanced MS/MS entropy similarities, particularly for spectra that were injected at low absolute quantities. Low abundant peaks represent the large majority of unknown compounds in metabolomic studies, rendering *Denoising Search* as a promising tool for major improvements in metabolome and exposome coverage.

We benchmarked our method against three alternative denoising algorithms. *Spectral Denoising* consistently outperformed the benchmarked alternatives, improving both the entropy similarity and the absolute quantity limit of high-confidence compound annotations. Despite varying levels of artificial noise, our method maintained robust performance. However, not all added noise ions were removed, primarily because our chemical plausibility checks were limited to the algorithms embedded in the Seven Golden Rules method[28], without considering molecular connectivity. Alternative approaches for recognizing true fragment ions involve the application of substructure annotation tools. Current software, such as Sirius[34], and MS-FINDER[36], often fails to recognize radical losses, which are prevalent in small molecule spectra (affecting over 60% of even-electron precursors spectra in NIST20)[30]. Without recognizing radical losses, true fragment ions may potentially be discarded. Therefore, subformula assignments that preserve valuable true fragment ions should be preferred over substructure annotation tools.

Yet, inherent limitations persist when solely relying on MS/MS spectra for compound annotation, even after removing noise ions. Reference spectra with inherently low spectral entropy are particularly vulnerable to noise ions, potentially leading to false negatives. The inability to differentiate between isomeric compounds and in-source fragments also presents significant challenges for the annotation of compounds in metabolomics when solely relying on MS/MS spectral matching. These observations indicate that employing hard thresholding based exclusively on spectral similarity is suboptimal for compound annotation in metabolomics and exposome research. Instead, *Denoising Search* should be supplemented with orthogonal experimental measures, such as retention time matching[37], molecular cross-section comparisons[38], and biological metadata screening[39], to further enhance the confidence of compound identifications in small molecule research. When applied to the latest ThermoFisher Scientific instrument, the Astral mass spectrometer, *Denoising Search* facilitated a 2.3-fold increase in the number of annotated compounds compared to classic MS/MS similarity investigations on an Exploris 240 mass spectrometer, with improvements noted across all seven major chemical superclasses.

Methods

Spectral Denoising

High-resolution mass spectra utilized for the development and validation of our *Spectral Denoising* algorithm were sourced from the licensed NIST23 Tandem Mass Spectral Library (2023 release). The explained ion intensity was calculated as the ratio of ion intensity retained post-denoising, denoted as, $I_{p,valid}$, to the total ion intensity of raw spectra, I_p, as demonstrated in equation (1):

$$Explained\ intensity\ (\%) = \frac{\sum_{p,valid} I_{p,valid}}{\sum_{p} I_p}\ (1)$$

Figure 1 (main text) visualizes the schema of the *Spectral Denoising* pipeline. All spectra were subjected to precursor removal before applying any form of *Spectral Denoising*, to ensure that residual intensities of the precursor ions do not inflate MS/MS matching scores.

Acquiring serial dilution MS/MS data of reference compounds to validate *Spectral Denoising*

Stock solutions of all target chemicals were prepared at 10 mM concentrations in methanol. Six mixtures of non-isobaric standards, each containing 40 compounds, were prepared by mixing 2.27 µL of each standard to achieve a concentration of 0.25 mM. 13 dilutions from these stock solutions were made to obtain final amounts to be injected into the LC-MS/MS systems, ranging from 0.02 pmol to 500 pmol: 0.02, 0.04, 0.1, 0.2, 0.4, 1, 2, 4, 10, 40, 100, 200 and 500 pmol (the concentration of the stock solution).

Prior to injections, solutions were dried and resuspended in 100 ul of the LC starting buffer. 2 ul volumes were injected onto a 10 cm, 2.1 mm i.d., 1.7 um particle Waters Acquity BEH amide column maintained at 30 °C with a flow rate of 0.4 mL/min, utilizing a gradient of mobile phases of water with 0.1% formic acid (A) and acetonitrile with 0.1% formic acid (B)[40]. Mass spectrometric detection was carried out on a Thermo Q- Exactive HF Orbitrap instrument (ThermoFisher Scientific, San Jose, CA) operated in positive electrospray ionization mode. Mass spectrometry was performed from a mass range 60-1500 m/z with a sheath gas flow rate 60, auxiliary gas flow rate 25, sweep gas flow rate 2, spray voltage 3.6 kV, capillary temperature 300 °C, S-lens RF level 50, and an auxiliary gas heater temperature 370 °C. MS^1 settings were set at a resolving power R=70,000, an automatic gain control target of 1e6, and a maximum injection time of 100 ms for single scans in centroid mode. MS/MS data were acquired in data-dependent mode at a resolving power R=15,000, an AGC target of 1e4, and a maximum injection time of 100 ms, with an isolation window of 1.0 m/z and no offset. The Top-N setting was 4, with an MSX count 1, loop count 4, and normalized collision energy steps 25, 35, and 65 NCE in centroid mode. For each mixture, precursor ions of the target compounds were specifically included for MS/MS acquisition as separate target inclusion lists to ensure that MS/MS spectra were acquired even for very low injected amounts, for which MS1 ion intensities may not have been found within the top-4 most abundant ions in an MS1 spectrum. Feature detection was performed on MS 4.9.2.

Benchmarking Spectral Denoising against alternative denoising algorithms

Algorithms were obtained from literature based on their premise to remove noise ions in MS/MS spectra and to promote spectral annotations. Methods were excluded if they relied on data integration across multiple spectra (‘consensus spectra’) or if they required auxiliary instrumentations[16, 17]. Three alternative denoising algorithms were implemented in Python 3.8. The reducing factor used for MS Reduce denoising was 90 with the maximum allowed quantization level of 11. For threshold denoising, the widely used 1% base peak height was selected as the predefined noise level. DNL denoising algorithm does not require additional parameter settings. The performance of the denoising algorithms was benchmarked on the 240 metabolite standards with absolute injected volumes from 200 pmol to 0.02 pmol, using the 500 pmol spectra as reference spectra. The improvements of MS/MS similarities for low abundant compounds were calculated using the ratio of the lowest injected quantity of compounds that yielded an MS/MS entropy similarity >0.75 of the raw spectra, divided by the lowest injected quantity of compounds that yielded an MS/MS entropy similarity >0.75 of the MS/MS spectra after the use of the benchmarked algorithms. A ratio less than 1 indicates that spectra gave <0.75 MS/MS similarity after denoising, even for the injected quantities of compounds for which raw spectra were annotated at MS/MS entropy similarity >0.75. Spectral entropy and entropy similarity were calculated as published before[21, 27].

Adding chemical and electronic noise ions to MS/MS spectra

To test the robustness of the *Spectral Denoising* method against three benchmarking algorithms, chemical noise and electronic noise were artificially added to all MS/MS spectra of the 240-compound mixtures with absolute injected volumes from 200 pmol to 0.02 pmol. The relative intensity of both electronic and chemical spectral noise I was generated using a Poisson distribution, demonstrated in equation (2):

$$f(I) = \frac{\lambda^I e^{-\lambda}}{I!} \quad (2)$$

Here, $f(I)$ represents the probability that a peak with relative intensity I will be generated, where $\lambda = 50$ characterizes the chemical noise and $\lambda = 5$ represents the electronic noise, to accurately mimic their respective behaviors. For chemical noise, *m/z* values were randomly selected from a database of 3.5 million formulas to ensure that only chemically feasible element ratios were used, with the additional constraint that noise ions did not exceed the precursor m/z. Electronic noise *m/z* values were randomly sampled from a uniform distribution ranging from 0 to the precursor ion m/z. If the calculated number of noise ions was not an integer, it was rounded up to the nearest integer to ensure that at least one noise ion was generated for each spectrum. The improvement in MS/MS entropy similarity scores was defined as the difference in entropy similarity between the 500 pmol reference spectra and the contaminated raw spectra or the 500 pmol reference spectra and the denoised spectra. The spectral entropy was calculated based on the 500 pmol reference spectra as the ground truth.

Acquiring and annotating HILIC-MS/MS data from plasma of AD patients using Orbitrap Exploris 240 and Orbitrap Astral mass analyzers with Denoising Search

The dataset comprised a subset of 20 plasma samples from an Alzheimer's patients exposome cohort, as part of an exploratory study coordinated by Duke University under Prof. Rima Kaddurah-Daouk. Samples underwent analysis using both the Orbitrap Exploris 240 and Orbitrap Astral systems (Thermo Scientific, San Jose). The same hydrophilic interaction liquid chromatography (HILIC) method was employed for both systems, employing a Waters ACQUITY Premier BEH Amide Column (1.7 µm, 2.1 mm x 50 mm). Gradient elution used a biphasic system consisting of (a) water and (B) 95% acetonitrile, both buffered with 10 mM ammonium formate and 0.125% formic acid. The gradient started at 100% phase B, reducing to 30% over 2.05 minutes, followed by an equilibration period back to 100% B over 0.65 minutes at 0.8 ml/min. For mass spectrometry, the Exploris 240 Orbitrap was set to perform an MS1 full scan (60-900 m/z range, 60,000 resolution, 1e6 AGC target, maximum injection time 100 ms) and a top-2 data-dependent MS/MS acquisition (DDA) (15,000 resolution, 1e5 AGC target, maximum injection time 10 ms, isolation window 1 m/z, normalized collision energies of 30-50-80%). The Astral system similarly conducted full scan MS1 (60-900 m/z range, 60,000 resolution, 1e6 AGC target, maximum injection time 100 ms) and MS/MS scans (15,000 resolution, 1% of 1e5 AGC target, maximum injection time 10 ms, isolation window 1 m/z, normalized collision energy of 40%). Cycle time was 0.2 msec, which in Astral was equivalent to approximately top-35 DDA-MS/MS. Electrospray ionization settings: spray voltage 3500 v (+), sheath gas 60 arbitrary units, axillary gas 20 arbitrary units, sweep gas 1 arbitrary unit, ion transfer tube temperature 350 °C, vaporizer temperature 400 °C, RF lens 50%. Plasma samples were extracted by a biphasic solution of MTBE/methanol/water as previously published[41], and aliquots were dried and resuspended in 100 µL of ACN:water (80:20) containing 30 isotope-labeled internal standards. 3 µl was injected. Pooled quality control samples, including reference material NIST SRM1950 plasma and blank quality controls, were analyzed to assess quantitative robustness and selectivity. Feature detection and alignment were performed using MS-DIAL (version 4.9.2). Compound annotations were performed using combined repositories of NIST23, Massbank.us, and GNPS libraries. Candidate spectra for identity search using entropy similarity and *Denoising Search* were restricted to a precursor ion mass tolerance of 10 mDa. Compound superclass information was assigned using the ClassyFire algorithm[42].

Data availability

NIST Tandem Mass Spectral Library, 2023 release (NIST23) spectra are commercial available and can be purchased from multiple vendors. MassBank of North America database (Massbank.us) spectra can be freely downloaded from Massbank.us (https://massbank.us/). The metabolome dataset of Alzheimer's Disease samples and the experimental data from the chemical dilution series can be requested from the authors. Source data are provided with this paper.

Code availability

The code for calculating spectral denoising and denoising search can be found at GitHub (https://github.com/FanzhouKong/spectral_denoising).

Acknowledgments

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Chapter 3: MolRex: Enabling Global Receptive Fields in GNNs for General Molecular Properties and Retention Time Prediction

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Abstract

Molecular property prediction is fundamental to computational chemistry and drug discovery, yet remains challenging due to the sparsity of molecular graphs and the locality of message passing in standard graph neural networks (GNNs). We present Molecular Representation Expansion (MolRex), a novel framework that augments sparse molecular graphs into fully connected graphs, equipping GNNs with a global receptive field. At the same time, MolRex preserves structural fidelity by encoding the original molecular topology through expressive, interaction-driven edge representations. Within this framework, we also propose Graph Isomorphism Attention Network (GIANT), a new GNN architecture that combines a revised multi-head attention mechanism from Transformer architecture with edge-aware aggregation from Graph Isomorphism Networks (GIN). We show that GIANT matches the expressiveness of the Weisfeiler–Lehman test and surpasses GAT/GATv2 baselines by introducing dedicated key matrices that explicitly incorporate edge encodings into message passing. Beyond molecular graphs, MolRex integrates molecular descriptors and fingerprints via a multi-stage encoding process, leveraging complementary information from traditional hand-engineered features. Extensive experiments demonstrate that MolRex consistently outperforms state-of-the-art methods, achieving median improvements of 9.7% reduction in RMSE across general molecular property benchmarks and 21.4% RMSE improvement on domain-specific tasks such as retention time prediction. Lastly, the pretrained MolRex was used to precompute retention times of >700,000 chemical

compounds from MS/MS reference spectral libraries. Results are freely available on MassWiki.metabolomics.us.

Introduction

Learning expressive and generalizable representations of small molecules is fundamental to numerous applications in chemistry and biology, including drug discovery[1], compound screening[2], and metabolite annotation[3]. Unlike proteins, which often consist of linear chains of recurring substructures, small molecules exhibit highly diverse topologies, many-body interactions, and complex intramolecular forces. These characteristics pose unique challenges for directly adapting widely used sequence-based frameworks such as transformers[4].

Graph Neural Networks (GNNs) have emerged as a natural choice for modeling molecular structures, where atoms are represented as nodes and chemical bonds as edges[5]. The Graph Isomorphism Network (GIN) remains a strong and widely used backbone in 2025, but its expressivity is fundamentally bounded by the Weisfeiler–Lehman (WL) test[6, 7]. Moreover, GIN treats all neighbors equally, preventing it from capturing nuanced intra-graph interactions. Graph Attention Networks (GATs) address this limitation by assigning different importance to neighbors via pairwise attention, borrowing ideas from transformers. However, they do not natively incorporate edge features—an essential aspect of chemical graphs [8, 9]—and thus remain WL-limited in expressive power. More broadly, all message-passing GNNs operating on sparse molecular graphs restrict information flow to k-hop neighborhoods, failing to capture long-range dependencies and implicit non-bonded interactions such as van der Waals and electrostatic forces.

Recent work has shown that transformers can be interpreted as a variant of GNNs when edge or spatial information is injected through attention biases or positional encodings[10]. The key advantage of transformers lies in their global receptive field, which naturally captures long-range dependencies between atoms. Methods such as Graphormer leverage shortest-path distance encodings and bond-aware attention to improve molecular modeling, achieving strong performance[11]. However, these approaches still constrain information flow through bond-defined topology. Moreover, transformers were originally designed for sequences, and their application to graphs often overlooks spatial priors that GNNs can exploit. Hybrid architectures such as GPS attempt to combine local message passing with global transformer encodings[12], but we find these approaches remain suboptimal compared to our method.

To overcome these limitations, we present Molecular Representation Expansion (MolRex), a foundation framework for small-molecule representation learning. MolRex augments sparse molecular graphs into fully connected graphs with expressive edge encodings that preserve original topology while equipping any GNN with a global receptive field. This design enables GNNs to capture both local structure and long-range dependencies without the computational overhead of full transformer models. Beyond graph structure, MolRex also unifies graph embeddings with QSAR features such as molecular descriptors and fingerprints, bridging domain-specific knowledge with learned representations.

Our contributions are threefold:

- Fully connected graph augmentation: a general mechanism that equips GNNs with a global receptive field, consistently boosting performance across architectures while capturing intramolecular interactions beyond chemical bonds.
- Graph Isomorphism Attention Network (GIANT): a novel message-passing network that extends GIN with multi-head attention while preserving WL-level expressiveness.
- Multi-modal molecular fusion: bridging domain-specific features with graph embeddings to achieve state-of-the-art performance on both general benchmarks and domain-specific tasks such as retention time prediction.

Related Works

Message Passing Neural Networks (MPNNs)

Message Passing Neural Networks (MPNNs) form the foundation of modern molecular graph learning. These models iteratively update node embeddings by aggregating and transforming information from neighboring nodes and edges[5]. Early variants such as Graph Convolutional Networks (GCN) and GraphSAGE pioneered neighborhood aggregation schemes, with GraphSAGE introducing learnable aggregation functions like mean, LSTM, and max pooling[13, 14]. The Graph Attention Network (GAT) further advanced this framework by computing attention coefficients to weigh the contributions of neighbors[8]. GATv2 extended this by allowing attention coefficients to depend on both source and target nodes symmetrically, improving expressivity[9]. The Graph Isomorphism Network (GIN) introduced a sum aggregation followed by a multilayer perceptron (MLP), achieving expressiveness equivalent to the Weisfeiler-Lehman (WL) test[7].

Later models, such as AttentiveFP, refined MPNN architectures for molecular property prediction by incorporating attention mechanisms at both atom and molecule levels while using GRUs to control the information change over multiple layers[15]. While these models have been successful in many small molecule tasks, they are still hindered by the information restricted to local sub-structure, because their receptive field is constrained to k-hop neighborhoods. These limitations motivate the exploration of more expressive architectures that go beyond the traditional message-passing paradigm.

Graph Transformers

Transformer architectures, originally developed for sequence modeling, have been successfully adapted for molecular graphs by employing special positional and spatial encoding for the graph topology. Unlike MPNNs, which rely on local aggregation within fixed neighborhoods, Transformers compute pair-wise interactions across all node pairs, making them well-suited for capturing long-range dependencies in molecular structures. More recently, graph-specific Transformers have emerged. Graphormer introduced structural priors such as shortest path distance and edge encoding into the attention bias, allowing the model to encode graph topology more effectively[11]. It also proposed a special centrality encoding mechanism to simulate the relative importance of nodes, which is vital in tasks involving quantum properties. These adaptations enable Graphormer to outperform traditional GNNs in many tasks, especially those requiring global context, such as toxicity prediction and quantum property regression.

Hybrid Architectures: GNN-Transformer Fusion

To harness both local inductive biases and global context modeling, recent research has proposed hybrid architectures that combine message passing and Transformer-based mechanisms. The *'General, Powerful, Scalable Graph Transformer'* (GPS) is a representative example of such integration. GPS combines a message-passing GNN layer for local structure modeling with a global attention layer, effectively decoupling local and global learning modules[12]. This separation enables each component to focus on the most relevant substructure. GNNs capture short-range, chemically relevant patterns, while Transformers learn long-range dependencies across the molecular graph.

Building on this framework, GPS++ introduces architectural refinements and integrates topological and geometric information through attention bias mechanisms[16]. For instance, spatial distances and shortest-path encodings are embedded as pairwise biases, allowing the attention mechanism to remain aware of graph topology and 3D conformation during training. These enhancements led GPS++ to achieve state-of-the-art results on OGB-LSC and PCQM4Mv2, especially in quantum chemistry prediction tasks.

Motif/Subgraph-Enhanced Graph Learning

Chemical substructures or motifs govern many molecular properties. Substructure examples include aromatic rings, functional groups, fused ring systems and many others. Incorporating this domain knowledge into graph neural networks has become a promising direction for improving representation quality. Both MGSSL (Motif-based Graph Self-Supervised Learning) and HiMol (Hierarchical Molecular Graph Self-Supervised Learning) introduce hierarchical modeling by decomposing molecules into atoms and motifs, with additional graph nodes attached[17, 18]. Through such graph augmentation, the augmented graph node can be used directly as graph representation. This architecture allows the model to reason about both fine-grained atomic interactions and coarse-grained chemical semantics.

Quantitative Structure-Activity Relationship Modeling

Before the rise of deep learning, Quantitative Structure-Activity Relationship (QSAR) models dominated the landscape of molecular property prediction[19]. These models rely on manually engineered features such as physiochemical descriptors, topological indices, and molecular fingerprints. Descriptors can include logP, molecular weight, polar surface area, and hydrogen bond donor/acceptor counts, often calculated using cheminformatics toolkits like RDKit[20] or Mordred[21]. Fingerprints, including ECFP (extended-connectivity fingerprints) and MACCS keys, represent molecular structures as binary vectors encoding the presence or absence of particular substructures[22, 23].

With the advent of deep learning, particularly graph-based architectures, the reliance on manual feature engineering has substantially decreased. Nevertheless, hand-crafted features remain highly relevant, both as standalone baselines and as complementary inputs in hybrid models. Notably, several recent approaches incorporate predictions from traditional models. An example is the use of random forests that were trained on molecular fingerprints as features to be used directly into the neural network's prediction head, resulting in significant performance gains[24]. These features, often derived from substructure fingerprints and physicochemical descriptors, can be viewed as distilled representations of domain knowledge. As such, they continue to provide strong inductive priors and play a crucial role in enhancing model robustness, especially in low-data or domain-specific regimes.

Methodology

MolRex

Figure 1a illustrates the overall architecture of MolRex, which is designed to learn expressive and hierarchical molecular representations through two complementary encoders. A graph encoder captures molecular topology, while a Dual Encoder processes molecular fingerprints and descriptors. Their outputs are fused via a gated feed-forward module consisting of two linear layers with a GELU activation in between, yielding the final molecular representation used for downstream prediction.

The graph encoder in MolRex employs a fixed-depth GNN backbone. In our case, we use a five-layer GIANT (Graph Isomorphism Attention Network), which will be described in detail in the following section. Unlike conventional molecular GNNs that operate on sparse graphs reflecting true molecular connectivity, and whose receptive field is restricted to K-hop neighborhoods, MolRex augments each molecule into a fully connected graph (Figure 1b). In this formulation, every atom is connected to all others, including itself, thereby enabling information flow beyond the K-hop limit and achieving a global receptive field.

A key drawback of this approach is the loss of explicit chemical connectivity defined by bonds. To address this, we design a specialized edge encoding scheme composed of four parts: (1) bond-end encoding, (2) bond-type encoding, (3) Relative Random Walk Probability (RRWP) encoding, and (4) 3D distance encoding. In chemistry, bond character is fundamentally determined by the polarity and electron distribution of the bonded atoms. Accordingly, bond-end encoding captures this information by embedding the atom types of both ends of the bond and summing them to ensure translational and rotational invariance. This design is more expressive than representing bonds solely by type (single, double, triple), since the bonded atoms directly influence bond polarity and strength. For instance, a C–C single bond, which is largely nonpolar, is chemically distinct from a C–O single bond, which is strongly polar due to uneven electron distribution. This contrasts with other GNN application domains, such as social networks, where edge semantics (e.g., a “follow” relation) are invariant to node identity. Bond type is additionally encoded with a separate embedding layer, while virtual edges introduced by graph augmentation (i.e., edges not corresponding to real chemical bonds) are assigned a special “no bond” token. The formulation of bond end and bond type encoding is given as (1) and (2):

$$\textit{Bond end encoding}: \mathrm{Emb}(\boldsymbol{x}_i) + \mathrm{Emb}\left(\boldsymbol{x}_j\right),\ d = 128\ (1)$$

$$\text{Bond type encoding: } \mathrm{Emb}\left(\boldsymbol{e}_{ij}\right),\ d = 128\ (2)$$

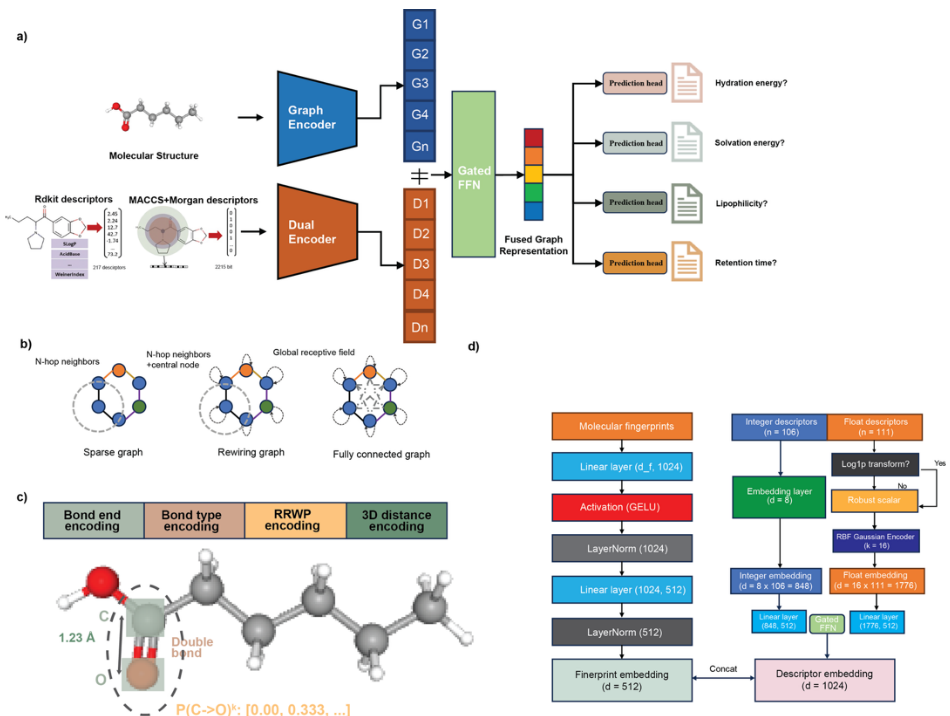

Figure 1. Overview of the MolRex architecture for molecular representation learning. (a) **MolRex architecture**. Molecular graphs are encoded with a GNN-based graph encoder, while molecular descriptors and fingerprints are processed by a Dual Encoder to generate complementary representations. The fused molecular features, obtained via a gated feed-forward layer (FFN), are then passed to task-specific prediction heads. (b) Graph connectivity. Comparison of sparse graphs, rewired graphs, and fully connected graphs. Sparse and rewired graphs restrict the receptive field to K-hop neighbors, whereas fully connected graphs enable global receptive fields. (c) Composition of edge encoding in MolRex, which consists of bond end encoding, bond type encoding, RRWP positional encoding and 3D distance encoding. (d) Dual Encoder design. The Dual Encoder consists of two branches: one for molecular fingerprints and one for molecular descriptors. The outputs are concatenated to form the combined descriptor–fingerprint embedding.

The RRWP encoding serves as a positional embedding that captures structural proximity through a restartable diffusion process. It is formally defined as:

$$X_{PE}^{k} = p(D, A, \mathrm{p\_int})\ (3)$$

$$\mathrm{RRWP(i)} = \mathrm{MLP}\left(\left[X_{PE}^1, X_{PE}^2, \ldots, X_{PE}^k\right]\right), d = 128 \; (4)$$

where X_{PE}^k denotes the positional encoding matrix at step k, $p(\cdot)$ represents the probability distribution obtained from a random walk with restart, D is the degree matrix, A is the adjacency matrix, and p_int is the initial distribution. A simple 2 layered MLP transformation is applied to the RRWP encoding to further maximize expressiveness[25]. Intuitively, RRWP encodes the probability that a node i reaches node j in k-steps, thereby quantifying how each node "sees" all others through a probabilistic diffusion process[26]. This can be viewed as a soft, probabilistic generalization of the shortest-path distance (SPD) encoding widely used in Graphormer[11], and a strict superset of the Random Walk Positional Encoding (RWPE), since RRWP reduces to RWPE when only the same start and end node are considered. In benchmarking by Grötschla *et al.*, RRWP outperformed alternative positional and spatial encodings, motivating its adoption in MolRex[25].

To complement structural encodings, we also incorporate 3D distance encoding. Atomic coordinates are first optimized using the Universal Force Field (UFF), and the lowest-energy conformer is selected. Pairwise interatomic distances are then calculated and used in place of raw coordinates to ensure translational and rotational invariance. This captures geometric relationships overlooked by purely bond-based topology. For example, in long-chain molecules, single bonds permit free rotation, and atoms many bonds apart can still be spatially close and chemically influential. Distances are expanded using K = 128 Gaussian kernel functions:

$$\forall i, j: \quad \psi_{ij}^k = \frac{1}{\sqrt{2\pi|\sigma^k|}} \exp\left(-\frac{1}{2}\left(\frac{|r_{ij}| - \mu^k}{|\sigma^k|}\right)^2\right) \in R \; (5)$$

where μ^k and σ^k are randomly initialized and optimized jointly with the GNN layers.

For descriptor and fingerprint encoding, MolRex employs a Dual Encoder to extract complementary domain-specific features. Descriptors are divided into two categories: (*i*) discrete integer descriptors (e.g., number of acidic groups) and (*ii*) continuous float descriptors (e.g., total surface area, molecular weight).

As illustrated in Figure 1d, integer descriptors are embedded directly, while float descriptors are preprocessed. Heavy-tailed distributions are identified by checking if the distance from the median to the 95th percentile exceeds three times the interquartile range (IQR), or if the 95th-to-median ratio exceeds three; descriptors meeting either condition are log1p-transformed prior to robust scaling. Normalized values are then encoded using K = 16 Gaussian kernel functions (as in Eq. 5). Both integer and float embeddings are projected to d = 512 through a feed-forward network and combined via gated concatenation to form the final descriptor embedding. For molecular fingerprints, MolRex encodes both Morgan fingerprints (capturing local substructures within radius 2) and MACCS fingerprints (capturing predefined functional groups) using a lightweight block of two linear layers, two GELU activations, and two LayerNorm layers, producing 512-dimensional embeddings. The final feature embedding is generated by concatenating descriptor and fingerprint embeddings, which are then fused with the graph embeddings for downstream tasks.

Graph Isomorphism Attention NetWork (GIANT)

The GIANT layer is a novel message-passing operator that integrates both attention-based aggregation and edge-conditioned representations, inspired by and extending GIN and Graph transformer. It operates on graphs *G = (V, E)* with node features $\boldsymbol{X} \in R^{N \times d}$ and edge features $\mathbf{E} \in R^{|E| \times f}$, and supports multi-head attention with H heads, just like the real transformers.

Each GIANT layer performs the following steps:

1. Linear Projections:

Input node features $\mathbf{x}_{\boldsymbol{i}}, \mathbf{x}j$ and edge encoding $\mathbf{e}ij$ are projected:

$$\boldsymbol{q_i} = W_q(\boldsymbol{x_i}) \ (6)$$

$$\boldsymbol{k_i} = W_k(\boldsymbol{x_i}) \ (\mathbf{7})$$

$$\boldsymbol{v_i} = W_v(\boldsymbol{x_i}) \ (8)$$

$$\boldsymbol{E}ij = \boldsymbol{W_{edge}}(\boldsymbol{e}ij) \ (9)$$

$$\boldsymbol{B}ij = \boldsymbol{B_{edge}}(\boldsymbol{e}ij) \ (10)$$

Where $\boldsymbol{W}_q$, $\boldsymbol{W}_k$, $\boldsymbol{W}_v$, $\boldsymbol{W}_{\boldsymbol{edge}}$ are all learnable weight matrices used to transform node embeddings and edge embeddings into the same latent space. $\boldsymbol{B}_{\boldsymbol{edge}}$ is also a learnable weight matrix but aims to produce attention bias per attention head from edge encoding. Unlike traditional GAT/GATv2, where the value weight matrix $\boldsymbol{W}_{\boldsymbol{v}}$ is omitted, MolRex provides this dedicated weight matrix to closely mimic the real transformer architecture. Furthermore, $\boldsymbol{E}ij$ is used as edge attribute in message passing process, just like GIN, while $\boldsymbol{B}ij$ is used as a special attention bias term when calculating attention coefficient computation process, just like other graph transformers. The edge information is injected into the GIANT twice to ensure the bond and spatial information is properly included and learned by neural network.

2. Attention Coefficient Computation:

Attention logits are computed as:

$$\alpha_{ij} = softmax\big(((\boldsymbol{q_i} + \boldsymbol{E}ij) * \boldsymbol{k_j^T})/\sqrt{d} + \boldsymbol{B}ij\big) \ (11)$$

where $\mathbf{q}_{\boldsymbol{i}}$ is the query matrix transformed receiving node, $\mathbf{k}j$ is the key matrix transformed sending node, $\sqrt{d}$ is the scale factor, and $\boldsymbol{B}ij$ is the attention bias term calculated per head.

3. Message Passing:

The message for node *i* is:

$$\mathbf{m}ij = \alpha ij \cdot (\mathbf{k}j + \mathbf{E}ij) \ (12)$$

4. Aggregation and update:

Aggregated messages are combined via:

$$\mathbf{h}i^{agg} = \sum j \in \mathcal{N}(i) \, \mathbf{m_{ij}} \ (13)$$

$$\mathbf{h}_{\boldsymbol{i}}' = MLP(\mathbf{h}_{\mathbf{i}}^{agg}) \ (\mathbf{14})$$

Residual connections are used to stabilize training and preserve gradient flow.

Expressivity Analysis of GIANT

We thoroughly analyzed the expressiveness of our GIANT with GAT and GIN under the Weisfeiler-Lehman (WL) test framework.

Comparison to GIN:

GIN is provably as powerful as the 1-WL test by design:

$$\mathbf{h}i' = MLP\left(\sum j \in \mathcal{N}(i)\,(\mathbf{k}j + \mathbf{e}ij)\right)\ (15)$$

GIANT generalizes GIN by incorporating learnable attention coefficients αij (Eq. 12), such that in special cases when αij is learned to be identical across all neighbor nodes and $\boldsymbol{W}_k$ and $\boldsymbol{W}_{edge}$ are learned to be identity operator, GIANT reduces to GIN-like layer. Thus, GIANT surpass GIN in expressivity and is at least as expressive as the WL test.

Comparison to Transformer:

By augmenting sparse molecular graphs into fully connected graphs, GIANT attains a true global receptive field, analogous to a standard Transformer layer. In a Transformer, the attention coefficient between nodes *i* and *j* is defined as:

$$\alpha_{ij} = \mathrm{softmax}\big((\mathbf{q}_{\boldsymbol{i}} * \boldsymbol{k}_{\boldsymbol{j}}^{\boldsymbol{T}})/\sqrt{d}\big)(16)$$

With an additional structural bias term, this becomes:

$$\alpha_{ij} = \mathrm{softmax}\big((\mathbf{q}_{\boldsymbol{i}} * \boldsymbol{k}_{\boldsymbol{j}}^{\boldsymbol{T}})/\sqrt{d} + \boldsymbol{B}ij\big)\ (17)$$

When edge attributes are ignored - or when the learned edge weight matrix W_{edge} converges to all zeros - the attention formulation in GIANT (Eq. 11) reduces exactly to Eq. 17, sharing the same bias term.

Furthermore, comparing to the update function in transformer:

$$\mathbf{h}_{\boldsymbol{i}}' = \big(\mathbf{h}_{\boldsymbol{i}}^{agg}\big)\ (\mathbf{18})$$

If the MLP used in GIANT's update function is learned to approximate the identity operator, then the GIANT update becomes identical to that of a graph Transformer.

In summary, under the conditions where (*i*) W_{edge} is all-zero, (*ii*) the MLP reduces to the identity, and (*iii*) both employ the same bias term, a GIANT layer reduces exactly to a graph Transformer layer while still maintaining a true global receptive field.

Experiments

Datasets

The benchmark includes six datasets that were used for two purposes: (1) predicting general molecular properties, and (2) predicting retention times (RT) under varying conditions of liquid chromatography methods. Each dataset is provided with predefined train/validation/test splits based on scaffold splitting, where molecules are partitioned according to their Bemis–Murcko scaffolds. Comparing to traditional random splits, the median Tanimoto-similarities between molecules in training and testing set has decreased by three-fold on most datasets. This setup introduces realistic distributional shifts and supports evaluation of out-of-distribution (OOD) generalization.

General molecular property regression tasks were tested on the ESOL, Freesolv and Lipo datasets from the MoleculeNet[27]. Specifically, the ESOL dataset contains experimental values for water solubility (log mol/L) for 1,128 small molecules. FreeSolv comprises 642 small molecules with experimentally measured and calculated hydration free energies in water (in kcal/mol). The Lipo (Lipophilicity) dataset contains experimental octanol/water distribution coefficients (logD at pH 7.4) for 4,200 compounds.

For domain-specific benchmarking, we evaluated three medium-sized retention time (RT) prediction datasets—AjsUoB, FemLong, and Eawag—all obtained from the RePoRT repository[28]. In addition, the METLIN SMRT dataset was included as a large-scale benchmark for evaluating model generalization[29]. Retention time reflects the duration that a molecule interacts with the stationary phase of a chromatographic column under a specific solvent gradient and buffer composition. Because these parameters vary across liquid chromatography (LC) methods, RT values are inherently method-dependent. Within a given LC method, however, retention behavior is largely governed by a compound's physicochemical properties—such as polarity, hydrophobicity, and ionization potential—making RT prediction a well-defined molecular property prediction task. The AjsUoB dataset comprises 947 RT measurements of small molecules analyzed using hydrophilic interaction liquid chromatography (HILIC). In contrast, both the FemLong and Eawag datasets are based on reversed-phase liquid chromatography (RPLC). FemLong features 415 small molecules analyzed using a long-gradient chromatographic setup, while Eawag includes 364 exposome-related compounds measured on an XBridge C18 column. Finally, the large-scale METLIN SMRT dataset encompasses 80,038 synthetic drug-like molecules, representing the most extensive RT collection currently available for supervised learning.

Benchmarks

Stage 1: Core GNN layers and graph augmentation.

We first evaluated the expressiveness and performance of the proposed GIANT convolutional layer, along with the effects of graph augmentation. This assessment was conducted not only on GIANT but also on two additional GNN baselines. We benchmarked against three widely used architectures: GAT, GATv2, and GIN. All models were constructed with five graph convolutional layers and employed identical node and edge encodings. The goals of this stage were twofold: (i) to assess the expressiveness and performance of the GIANT convolutional layer in isolation, and (ii) to evaluate the performance gains achieved by equipping any GNN with a global receptive field through graph augmentation.

Stage 2: Dual Encoder vs. naïve chemical feature encoders and classical models.

To fully leverage the expressive power of neural networks, we implemented a carefully designed encoding framework to preprocess molecular descriptors and fingerprints (see Methodology). We compared the performance of our Dual Encoder against simpler baselines, including a naïve linear projection of chemical features and traditional machine learning models such as random forests. This experiment highlights the importance of robust feature preprocessing and encoding for molecular property prediction.

Stage 3: General-purpose molecular property prediction.

We next evaluated the complete MolRex architecture on three widely used molecular property prediction datasets: ESOL, FreeSolv, and Lipophilicity (Lipo). We compared MolRex against a suite of strong baselines, including state-of-the-art architectures such as Graphormer, GPS, AttentiveFP, HiMol, and Pretrained GIN. This stage aimed to assess MolRex's generalizability across diverse molecular property prediction tasks.

Stage 4: Domain-specific retention time prediction.

We further benchmarked MolRex on the specialized task of retention time (RT) prediction. Baselines included RT-Transformer, a state-of-the-art method employing stacked GAT layers with residual connections and 1D convolutional layers for fingerprint encoding, as well as ReTip 2.0, a pure descriptor-based machine learning model built on AutoGluon. RT prediction is known to depend on subtle, context-dependent molecular properties; thus, this evaluation demonstrates the adaptability and robustness of MolRex in experimentally grounded, domain-specific tasks.

Stage 5: Application to metabolite annotation.

Finally, we applied MolRex to public UC Davis metabolite datasets, available in MassWiki. The model was trained on two chromatographic systems - HILIC and reversed-phase (RP) - and evaluated using sets of true positive and true negative annotations. Beyond performance benchmarking, MolRex was further employed to flag and exclude probable false annotations within existing datasets. To this end, we curated a dedicated true negative dataset comprising known misannotations, identified based on implausible chromatographic or biological evidence such as poor or absent peak shapes, known contaminants, or inconsistent biological context. These compounds were initially annotated by MS/MS spectral matches with high-entropy similarity scores yet lacked credible chromatographic support. In total, the HILIC dataset contained 126 true negative compounds, while the RP dataset included 30 misannotations. These datasets were used to assess MolRex's ability to identify and eliminate incorrect annotations based on predicted retention times, thereby demonstrating its practical utility in enhancing metabolite annotation workflows.

Implementation details

All experiments were conducted on UC Davis' HIVE computing cluster equipped with NVIDIA A6000 GPUs. Our implementation stack included Python 3.12, CUDA 12.1, PyTorch 2.7.0, and PyTorch Geometric 2.6.1. Molecular, atomic, and bond-level features were extracted using RDKit version 2024.09.4. Unless otherwise specified, all models used a hidden dimension of 512. The prediction head for every model was implemented as a two-layer multilayer perceptron (MLP) with GELU activation between layers. All models were trained using the AdamW optimizer, and the maximum number of training epochs was set to 500. Hyperparameters such as learning rate and number of readout layers were tuned using the Optuna framework with 20 trials per model. Each trial was trained for up to 100 epochs, and the configuration achieving the lowest validation loss was selected for final evaluation. No dropout was used as, because dropout does not prevent overfitting while introducing uncertainty in regression tasks[30]. The best-performing checkpoint, as determined by the lowest validation loss, was used for reporting test set performance.

Results

Table 1 summarizes the performance of three GNN variants and the effect of graph augmentation. When evaluating convolutional layers in isolation, GIANT consistently outperforms the two baseline GNNs in both in-distribution error (validation RMSE) and out-of-distribution error (test RMSE). We also report balanced RMSE, defined as the mean of validation and test RMSE, where RMSE is a commonly used metric to describe the model performance by measuring the average deviation between predicted and experimental results, expressed in corresponding unit. Overall, GIANT with augmented graphs achieves the best results in 11 out of 12 metrics. The only exception is the FreeSolv test set, where GIANT with sparse graphs performs slightly better; however, the augmented GIANT variant still outperforms all baseline GNNs (both sparse and augmented). On average, GIANT with graph augmentation delivers a median performance gain of 5% compared to other methods, with the largest improvement (12.8%) observed on balanced RMSE for the FreeSolv dataset. These findings demonstrate that (*i*) our novel GIANT convolutional layer with fully connected graphs and global receptive fields surpasses traditional GNN baselines, and (*ii*) even baseline GNNs benefit from graph augmentation, confirming the effectiveness of enabling global receptive fields.

Table 2 highlights the performance advantages of our Dual Encoder compared to simpler baselines, including random forests and naïve MLPs. Naïve MLPs perform poorly because most neural network architectures implicitly assume features of comparable magnitude and approximately normal distributions. Molecular descriptors, however, often vary widely in scale and exhibit heavy-tailed distributions, which severely degrades naïve MLP performance. Random forests, in contrast, do not assume normality and are relatively robust to differences in feature magnitude due to their decision tree–based ranking mechanism, yielding stronger results than naïve MLPs. With proper preprocessing and encoding, our Dual Encoder consistently outperforms both baselines, achieving a median 11% performance gain across six datasets. The most notable improvement was observed on the AjsUob dataset, where Dual Encoder reduced RMSE by nearly 50% compared to the strongest baseline.

Moving beyond individual building blocks of MolRex, we further benchmarked the MolRex models against several well-established molecular representation learning methods, including the pure graph Transformer (Graphormer), the hybrid Transformer-GNN architecture (GPS), the motif-enhanced model (HiMol), the attention-based AttentiveFP, pretrained GIN, and classic machine learning model random forest with descriptor/fingerprints as feature sets (Table 3). Across all three molecular property regression tasks, MolRex consistently outperformed all baseline state-of-the art models (SOTA), achieving improvements ranging from 5.6% to 18.8% compared to SOTA methods.

Table 1: Benchmarking performance of GIN layers and graph augmentations. Bold font denotes the best performances. All results are repeated 5 times and reported the averages. The GNN with appendix _expanded indicates GNNs equipped with augmented graph thus enabling a global receptive field. The RMSE values have the same unit as the experimental measurements.

Data	Esol (log mol/L)			Freesolv (kcal/mol)			Lipo (logD, dimensionless)		
RMSE	Validation	Test	Balanced	Validation	Test	Balanced	Validation	Test	Balanced
GIN	0.863	0.9794	0.863	2.166	2.337	2.251	0.671	0.722	0.696
GATv2	0.959	0.907	0.933	2.133	2.579	2.356	0.678	0.735	0.707
GIANT	0.855	0.890	0.853	2.827	**1.674**	2.251	0.636	0.714	0.675
GIN_expanded	0.949	0.912	0.931	2.467	1.770	2.119	0.671	0.722	0.696
GATv2_expanded	0.842	0.881	0.862	2.163	2.254	2.209	0.708	0.763	0.736
GIANT_expanded	**0.808**	**0.876**	**0.842**	**1.911**	1.787	**1.849**	**0.622**	**0.696**	**0.659**

Table 2: Benchmarking performance of Dual Encoder against a Random Forest model and a naïve MLP model. Bold font denotes the best performances. All results are repeated five times and report the averages. N/A values indicate exploding gradients (where the gradients of the loss function become excessively large, leading to unstable training), and loss values become +inf. The RMSE values have the same unit as the experimental measurements.

Data Test RMSE	Esol (log mol/L)	Freesolv (kcal/mol)	Lipo (logD, dimensionless)	Ajsuob (min)	Eawag (min)	Femlong (min)
Random Forest	0.983	2.440	0.780	2.061	2.100	7.778
Naïve MLP	1.078	3.874	N/A	1.220	4.266	N/A
Dual Encoder	**0.887**	**1.934**	**0.688**	**1.017**	**1.638**	**5.770**

Table 3: Benchmarking MolRex against other molecular property prediction models. Bold fonts denote the best performances. Results report the averages of five repeats. The RMSE values have the same unit as the experimental measurements.

Data Test RMSE	Esol (log mol/L)	Freesolv (kcal/mol)	Lipo (logD, dimensionless)
Graphormer	1.096	1.943	0.846
GPS	0.884	4.043	0.735
HiMol	0.943	2.266	0.748
AttentiveFP	0.969	2.09	0.739
GIN_pretrained	1.12	2.817	0.729
Descriptor+Random Forest	0.883	2.440	0.780
MolRex	**0.833**	**1.579**	**0.658**

We next evaluated MolRex on retention time (RT) prediction under three experimental conditions (Table 4). MolRex outperformed all baseline models across the three datasets, achieving substantial improvements of 18.9% on AjsUoB, 21.5% on Eawag, 21.3% on Femlong, and 7.5% on METLIN SMRT dataset, compared to current state-of-the-art methods. Interestingly, METLIN DLM, a purely descriptor/fingerprint based multilayer perceptron model, consistently outperformed the more complex RT-Transformer, which relies on stacked GAT layers and convolutional fingerprint encoders. This finding highlights that domain-specific chemical descriptors remain highly informative and cannot be disregarded. Nevertheless, MolRex consistently surpassed both descriptor-based and GNN-based baselines, demonstrating its ability to generalize effectively across both general-purpose and domain-specific molecular property prediction tasks.

Table 4: Benchmarking MolRex against other retention time prediction models. Bold fonts denote the best performances. Results report the averages of five repeats. The METLIN SMRT dataset is a large-scale dataset, thus the results are highlighted in orange. The RMSE values reported here are all in minutes.

Data Test RMSE	Ajsuob (min)	Eawag (min)	Femlong (min)	METLIN SMRT (min)
RT-Transformer	1.97	4.82	18.9	1.52
METLIN DLM	1.17	2.18	7.09	1.46
Retip 2.0	1.78	2.62	7.82	3.41
MolRex	**0.95**	**1.71**	**5.58**	**1.35**

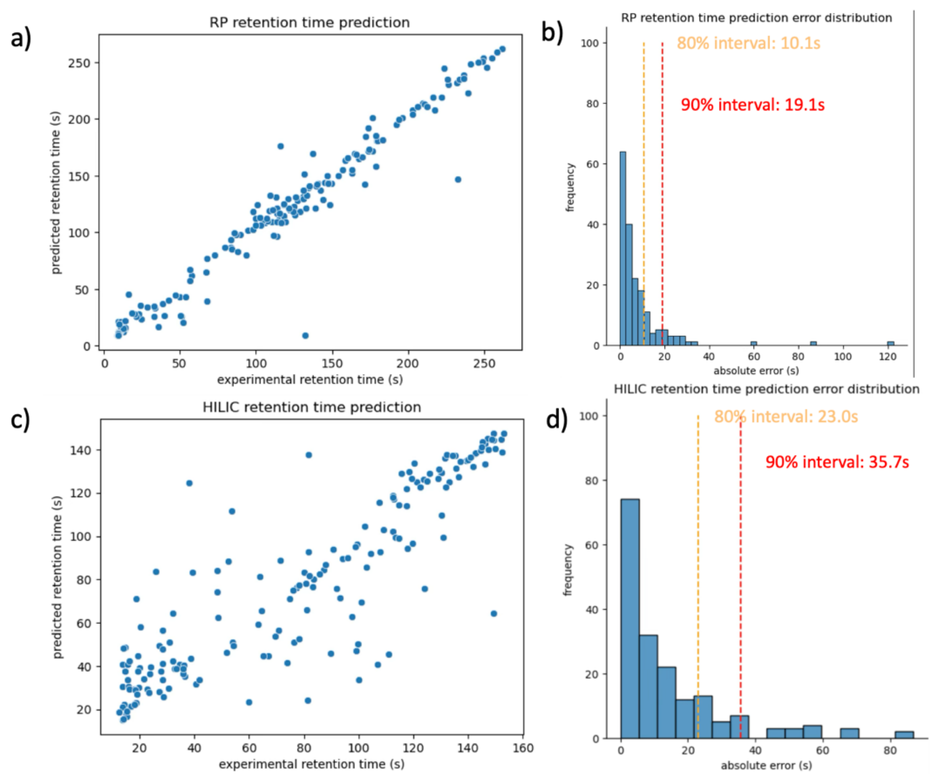

Figure 2. Performance of MolRex for chromatographic retention time prediction across two UC Davis metabolomics retention time datasets. (a) Predicted versus experimental retention times for the reversed phase (RP)dataset. (b) Distribution of absolute prediction errors for the RP dataset, with the 80% and 90% confidence intervals marked at 10.1 s and 19.1 s, respectively. (c) Predicted versus experimental retention times for the hydrophilic interaction chromatography (HILIC) dataset. (d) Corresponding error distribution for HILIC predictions, indicating that 80% and 90% of predictions fall within 23.0 s and 35.7 s, respectively.

Lastly, we applied MolRex to two UC Davis metabolomics retention time (RT) datasets, available for downloads from MassWiki.metabolomics.us. Data was acquired using hydrophilic interaction liquid chromatography (HILIC) and reversed-phase (RP) columns, and compounds were manually annotated based on accurate mass and MS/MS similarity from biological samples. Hence, it was expected that the datasets contained multiple false hits. To perform model testing on likely true positive hits, rigorous quality filtering was performed to ensure the reliability of the training, validation, and testing sets. Specifically, data from positive and negative electrospray ionization (ESI) modes were combined, and the following filters were applied:

(a) compounds exhibiting multiple adduct annotations within a 5 s normalized RT window were retained to improve annotation consistency; and

(b) spectra were required to have spectral entropy > 0.5 and normalized entropy < 0.8, thereby excluding low-information or excessively noisy spectra from the training process.

After applying these coarse filtration criteria, we obtained an RP dataset containing 1,701 annotated compounds and a HILIC dataset containing 1,851 annotated compounds, each representing compound annotations suitable for supervised RT prediction.

Figure 2 summarizes the performance of MolRex in predicting chromatographic retention times for these datasets. Panels a and c illustrate the correlations between predicted and experimentally measured retention times for the RP and HILIC datasets, respectively. For the RP dataset, MolRex achieved $R^2 = 0.95$, RMSE = 15.36s, MAE = 8.21s, and a median absolute error = 4.67s, demonstrating excellent agreement between predicted and observed retention behavior. The HILIC model yielded $R^2 = 0.77$, RMSE = 21.78s, MAE = 14.44s, and a median absolute error = 7.49s, reflecting slightly greater dispersion consistent with the higher physicochemical heterogeneity inherent to HILIC separations. Panels b and d present the corresponding prediction-error distributions. In both cases, the majority of predictions fall within narrow confidence intervals. 80 % of the predictions were found within 10.1s for RP and 23.0s for HILIC, and 90 % within 19.1s and 25.7s, respectively, indicating that MolRex provides highly consistent retention time estimates across chromatographic modalities.

To further evaluate the practical utility of these models, the trained models were applied to curated misannotation (true-negative) datasets to assess their capability to flag false identifications using predicted RTs, even when spectral similarity scores were high. On the RP true-negative dataset, 91.4 % of misannotated compounds were correctly flagged using the 80 % prediction-error quantile (10.1s). For the HILIC dataset, 71.3 % of misannotations were excluded using the corresponding 80 % quantile (23s), demonstrating that retention-time prediction can serve as an effective orthogonal filter for annotation quality control. Finally, the optimized MolRex models were applied to pre-compute retention times for over 700,000 compounds from available reference libraries (MassBank.us, GNPS, NIST23). These predicted retention times are publicly accessible via MassWiki.metabolomics.us to facilitate future metabolite annotation and benchmarking studies.

Discussion

The results presented in this chapter demonstrate that MolRex, a foundation-style molecular representation learning framework, substantially improves chromatographic retention time (RT) prediction relative to existing approaches. By augmenting molecular graphs with global connectivity, incorporating 3D distance-aware encodings, and fusing learned graph embeddings with molecular descriptors and fingerprints, MolRex addresses fundamental limitations of message-passing graph neural networks (GNNs).

These findings align with recent advances in molecular machine learning that emphasize the need for global context and hybrid feature integration. Traditional quantitative structure–retention relationship (QSRR) models relied on hand-crafted descriptors coupled with regression[31, 32], providing interpretable yet limited performance across diverse chromatographic systems. More recent GNN-based methods[7, 33] introduced structural awareness but remained constrained by Weisfeiler–Lehman (WL) equivalence limitations, which restrict the ability to capture stereochemistry and long-range dependencies[7, 33]. Transformer-inspired architectures[11, 26] have begun to overcome these bottlenecks by encoding pairwise distances and positional information, yet most implementations have not explicitly addressed retention time prediction in metabolomics. By combining descriptor/fingerprint fusion with distance-aware attention, MolRex bridges these gaps and achieves state-of-the-art performance across LC modalities.

However, the reduced predictive accuracy observed for the HILIC dataset highlights the intrinsic complexity of HILIC retention mechanisms relative to reversed-phase (RP) chromatography. Unlike RP systems, dominated largely by hydrophobic partitioning, HILIC retention arises from a multifactorial interplay of partitioning, adsorption, electrostatic, and hydrogen-bonding interactions between solutes and the highly hydrated stationary phase[34]. These mechanisms are strongly influenced by solvent composition, water-layer dynamics, and analyte ionization states, leading to nonlinear and context-specific retention behaviors. Consequently, molecules with intermediate polarity ("medium-retained" compounds) often occupy a transition zone between hydrophilic partitioning and electrostatic adsorption, where small perturbations in molecular structure (e.g., pKa shifts, tautomerism, or conformational changes) can cause disproportionately large retention shifts[35-37]. This mechanistic heterogeneity makes it particularly challenging for data-driven models to learn a consistent mapping between molecular features and retention time. Subsequent research will individually test the outliers found for both RP- and HILIC models using authentic chemical standards; yet, these validations were outside the scope of the cheminformatic nature of this chapter. Similarly, future research may devise mechanisms to better understand and compensate for the intricate retention mechanisms in HILIC methods. Possibly, chemical descriptor fingerprints may be better suited to capture this type of domain-specific knowledge than purely graph-based methods.

From a machine learning standpoint, such diversity in retention mechanisms effectively increases the dimensionality of the feature–property space, requiring substantially larger training sets to achieve comparable generalization to RP systems. Empirical learning-curve analyses in chromatographic modeling suggest that achieving RP-level performance ($R^2 \approx 0.9$) for HILIC may require 5–10× more high-quality, method-consistent data, likely on the order of 10^4 unique compounds encompassing diverse ionizable and zwitterionic chemotypes[38, 39]. Expanding annotated datasets to include more polar, amphoteric, and ionic species, along with explicit method descriptors (e.g., buffer type, salt concentration, pH), could substantially reduce prediction variance. Incorporating mechanistic priors such as charge-aware embeddings, solvent-accessible surface area descriptors, and simulated partition coefficients may further enhance the model's capacity to capture mixed-mode interactions. Ultimately, improving HILIC retention prediction will depend on both larger, more chemically representative datasets and architectures capable of disentangling overlapping retention mechanisms, enabling AI-based approaches to more faithfully model the continuum between hydrophilic and electrostatic retention.

Importantly, the integration of RT as an orthogonal feature enhances confidence in metabolite annotation, complementing spectral similarity-based identification. This observation supports prior reports that accurate RT prediction can improve annotation pipelines in untargeted metabolomics[40, 41], and recent community guidelines emphasize RT as a key orthogonal criterion for Level 1–2 metabolite identification[42, 43]. By achieving improved predictive accuracy across multiple liquid chromatography setups, MolRex contributes to resolving the long-standing challenge of QSRR transferability across chromatographic platforms, a problem highlighted in comparative studies[44].

The demonstrated generalizability of MolRex further suggests potential for broader applications beyond RT prediction. Analogous to foundation models in natural language processing and protein folding[45], MolRex can serve as a general-purpose molecular encoder for tasks ranging from bioactivity prediction to exposome annotation, where small molecules of diverse origins must be reliably identified and characterized. The inclusion of hybrid molecular features resonates with recent findings that descriptor/graph fusion consistently outperforms unimodal approaches[46] .

Finally, by explicitly integrating both local and global structural information, MolRex contributes to ongoing discussions in the cheminformatics community regarding the optimal balance between chemically intuitive features and data-driven representations. Whereas purely data-driven models risk overfitting to narrow training distributions, and descriptor-based models often lack sufficient expressivity, MolRex provides a principled middle ground that leverages the strengths of both paradigms. Nevertheless, like most graph-based molecular learning frameworks, MolRex inherits an important limitation: its inability to fully account for tautomerism. Tautomers are graphically distinct but chemically equivalent species that coexist in dynamic equilibrium, particularly under varying pH or solvent conditions. Because GNNs treat molecular graphs as static topologies, they cannot inherently recognize these interchangeable forms as the same chemical entity. As a result, tautomeric shifts can lead to inconsistent embeddings and reduced predictive accuracy for molecules whose predominant form varies across chromatographic or ionization conditions. Addressing this limitation may require future extensions that incorporate tautomer-aware molecular representations, ensemble graph encodings, or probabilistic conformer sampling to more faithfully capture the dynamic chemical landscape underlying experimental measurements.

Conclusion

In this study, we introduced MolRex, a comprehensive framework for molecular representation learning with several key contributions. First, we developed GIANT, a novel message passing architecture that surpasses widely used baselines such as GATv2 and GIN in expressive power and is provably at least as powerful as the first-order Weisfeiler–Lehman (WL) test. Second, we demonstrated that global context can be incorporated through dense graph connectivity: by expanding sparse molecular graphs into fully connected graphs with carefully designed edge encodings, we achieve a global receptive field while preserving the semantics of chemical bonds. Third, we established best practices for encoding cheminformatics features, systematically evaluating and identifying effective strategies for embedding molecular descriptors and fingerprints into deep learning architectures.

Beyond these methodological advances, MolRex enables two important practical outcomes. It provides a robust tool to automatically predict retention times for any LC–MS method, thereby offering a scalable solution for chromatographic modeling. Moreover, MolRex empowers researchers to validate or invalidate metabolite annotation claims, including those from legacy datasets and prior studies, by providing an orthogonal check against spectral similarity–based identification. Together, these contributions position MolRex as a versatile and powerful framework that bridges methodological innovation in graph representation learning with practical impact in metabolomics and small-molecule analysis.

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Conflict of Interest

None declared.

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Overall discussion

Over past years, metabolomics has evolved into a pivotal discipline for understanding the chemical basis of biology and disease. By quantitatively profiling small molecules that arise from enzymatic activity and environmental interaction, metabolomics reveals the dynamic biochemical state of living systems in ways that genomics or proteomics alone cannot[1]. Its capacity to capture both endogenous metabolism and exogenous exposures makes it a foundation of exposome research, linking molecular processes to lifestyle, environment, and health outcomes[2, 3]. Advances in high-resolution liquid chromatography–tandem mass spectrometry (LC–MS/MS) have dramatically expanded metabolite coverage, enabling simultaneous measurement of thousands of features across complex biological matrices[4]. However, substantial challenges remain, such that spectral libraries still represent only a fraction of known chemical diversity[5], noise and co-fragmentation continue to obscure true fragment ions[6, 7], and current machine learning models struggle to fully encode the structural topology and physicochemical complexity of small molecules[8]. These limitations underscore the need for new data curation strategies, novel data quality improvement approaches, and representation learning frameworks capable of capturing the full analytical and chemical diversity of the metabolome.

This dissertation addresses these challenges through three synergistic developments—LibGen, *Spectral Denoising*, and MolRex—that collectively enhance the reproducibility, sensitivity, and scalability of computational metabolomics. LibGen automates the generation and curation of high-quality reference libraries, enabling systematic and consistent spectral organization. *Spectral Denoising* enhances annotation sensitivity and reliability in low-signal regimes by removing noise ions and therefore improving the quality and interpretability of experimental MS/MS spectra. MolRex, a foundation-style molecular representation learning framework, achieves highly accurate retention time prediction and demonstrates strong generalizability across diverse chromatographic systems and general molecular property prediction tasks. Together, these advances address three critical components of metabolite annotation: (i) producing comprehensive and reliable reference spectra, (ii) improving the quality of experimental data to minimize noise and ambiguity, and (iii) increasing annotation confidence by integrating predicted and observed retention times as orthogonal, corroborative evidence.

The development of LibGen marks a key advance in addressing the long-standing challenge of scalable, high-quality spectral resources. Manual curation in major repositories such as NIST[9], METLIN[10], and MassBank.us[11] is time-intensive and inconsistent, while community-driven databases like GNPS[12] depend on variable user-submitted data. LibGen integrates chemistry-aware denoising, mass error correction, and fragment validation-based spectral quality assessment to automate and standardize library building. Applied to ultraviolet photodissociation (UVPD), electron-activated dissociation (EAD), and higher-energy collisional dissociation (HCD) mass analyzers, LibGen produced the first open-access UVPD and EAD spectral libraries for natural products with 1,093 and 2,241 compounds, broadening coverage beyond drug-like molecules that dominate current databases, with extremely diverse chemical coverage. These resources enable the creation of richer spectral foundations analogous to peptide reference libraries in proteomics, providing a data infrastructure upon which future machine-learning and spectral-prediction tools can be trained.

The *Spectral Denoising* framework directly addresses one of the persistent challenges in metabolomics—the pervasive presence of noise ions in MS/MS spectra—which continues to impede confident compound annotation in exposome and clinical studies. Random noise and ion co-fragmentation frequently generate *chimeric spectra* that obscure genuine fragment ions[13], particularly for xenobiotics and low-abundance metabolites near the limits of quantification[14, 15]. By integrating subformula plausibility analysis[16] with intensity frequency modeling, *Spectral Denoising* effectively distinguishes authentic chemical signals from noise fragment ions, enhancing spectral quality while preserving informative low-intensity ions typically lost to conventional filters. Across both experimental and synthetic datasets, *Spectral Denoising* demonstrated superior performance compared with four widely used denoising methods, underscoring its robustness across varying noise levels and fragmentation conditions. When integrated with entropy similarity, the method was extended into a unified framework termed *Denoising Search*, which increased the number of confidently annotated compounds by 25%–53% across two biological datasets without elevating false-positive rates. The observation here is consistent with reports that robust noise handling can uncover hidden low-abundance metabolites[17]. This capability is particularly impactful for exposome-level metabolomics, where weak spectral signals often correspond to environmental contaminants, dietary constituents, or xenobiotic metabolites[18-20]. By recovering chemically plausible spectra from data previously dominated by noise*, Spectral Denoising* transforms "dark matter" in metabolomics[21] into interpretable information, expanding the detectable chemical landscape and facilitating the discovery of novel biomarkers of environmental and nutritional exposure.

The third pillar, MolRex, introduces a foundation-style molecular representation learning framework designed for both chromatographic retention time as well as general molecular property predictions. Retention time serves as critical role in boosting annotation confidence by providing orthogonal and complementary information beyond traditional MS/MS spectral matching. Traditional quantitative structure–retention relationship (QSRR) models depend on manually engineered descriptors and linear regression[22], while more recent deep learning approaches—such as graph neural networks (GNNs)—offer improved expressivity but remain limited by local message passing and a lack of global chemical context[23, 24]. MolRex bridges this gap by integrating graph-level embeddings, 3D distance encodings, and physicochemical descriptors through a gated fusion mechanism, thereby capturing both local structural details and long-range interactions. Moreover, by augmenting sparse molecular graphs into fully connected graphs with edge-aware encodings, MolRex enables a global receptive field for any GNN backbone, overcoming one of the fundamental limitations of conventional message-passing architectures. This hybrid design enhances prediction accuracy, robustness, and cross-domain transferability, positioning MolRex as a molecular analogue to large foundation models in other domains—capable of few-shot and even zero-shot generalization to novel chemical structures. Building on this strong predictive performance, the pretrained MolRex model was applied to predict retention times for approximately 700,000 molecules across two in-house chromatographic systems. These large-scale predictions provide a valuable resource for future studies, serving as a foundation for confidence score modeling and as a complementary feature to other annotation frameworks aimed at improving the accuracy and confidence of compound identification.

Beyond these specific contributions, the broader metabolomics field is rapidly converging with artificial intelligence (AI) and foundation-model development. New large-scale tools such as DreaMS (Deep Representation learning of Mass Spectra) have demonstrated that self-supervised training on millions of unlabeled tandem MS spectra can yield universal spectral embeddings that enable structure prediction, spectral similarity search, and cross-instrument transfer learning[25]. These spectral encoders parallel the evolution of protein language models (e.g., ESM, ProtT5)[26, 27] and large molecular transformers (e.g., ChemBERTa, Graphormer)[28, 29], reflecting a shift from handcrafted spectral similarity measures to learned representations that capture the underlying chemistry of fragmentation. The integration of such spectral foundation models with curated, high-quality datasets—such as those generated through LibGen—will be critical for improving spectral interpretation and bridging the gap between raw spectra and molecular structure. Similarly, AI-driven methods are redefining how spectral annotation, compound identification, and property prediction are performed. Tools such as CFM-ID[30], NEIMS[31], and MS2Deepscore[32] now use neural networks to simulate fragmentation or learn spectrum–structure embeddings directly, while models like DreaMS[25] and Spec2Mol[33] extend these capabilities through contrastive or generative pretraining. The increasing availability of large, standardized datasets will enable hybrid pipelines where denoising algorithms like *Spectral Denoising* provide clean input data for pretrained spectral encoders, substantially improving model accuracy and transferability. Furthermore, the combination of spectral and molecular embeddings—such as integrating DreaMS-style encoders with MolRex representations—could lead to unified molecular-spectral foundation models capable of learning across modalities, mirroring multimodal AI systems in computer vision and natural language processing.

While these developments are highly promising, several limitations and open questions remain. High-quality training data are still scarce for many ionization and fragmentation modes, particularly for natural products and environmental xenobiotics, which are often underrepresented in public spectral libraries[34]. As a result, AI models trained on narrow chemical distributions may fail to generalize to structurally diverse or uncommon compound classes[35]. Similarly, spectral denoising and encoding approaches must carefully balance sensitivity and specificity to avoid amplifying background noise or introducing bias toward abundant species. Although false discovery rate (FDR) analyses in this work demonstrate strong performance, further evaluation across a broader spectrum of compound classes would be valuable for defining the true operational boundaries of these methods. For molecular representation learning, tautomerism remains a key challenge: tautomers are thermodynamically interconvertible yet differ in bonding topology, leading graph-based models to treat them as distinct entities. Addressing this discrepancy will likely require probabilistic conformational models capable of predicting tautomeric distributions under specific experimental or chromatographic conditions. Ultimately, overcoming these challenges will depend on continued integration of analytical chemistry expertise with machine learning, emphasizing interpretability, uncertainty quantification, and open data sharing to ensure robust and generalizable progress in metabolomics.

Looking ahead, the convergence of curated data resources, denoising algorithms, and foundation-style AI models promises to reshape metabolomics into a more predictive and mechanistically grounded science. Integration of LibGen-curated spectral libraries with DreaMS-like encoders could enable large-scale spectral pretraining, analogous to the evolution of genomics into the era of large foundation models. *Spectral Denoising*, in turn, can serve as a preprocessing layer to enhance spectral clarity for downstream AI tasks. MolRex provides the deep learning backbone for property and bioactivity prediction, with potential extensions toward joint molecular–spectral representation learning. Together, these advances outline a path toward self-improving metabolomics systems, where analytical data continuously refine AI models that, in turn, enhance data interpretation and annotation.

In conclusion, this dissertation demonstrates that advancing metabolomics requires progress at the intersection of analytical chemistry, computational modeling, and AI-driven data science. By automating spectral curation, improving low-abundance annotation, and developing generalizable molecular representations, the presented work lays essential groundwork for the next generation of machine-learning–powered metabolomics. Future research integrating spectral foundation models like DreaMS with domain-specific tools such as LibGen, *Spectral Denoising*, and MolRex may ultimately yield an end-to-end learning framework capable of transforming raw spectra into chemical and biological insight—realizing the long-envisioned goal of a fully predictive, data-driven metabolomics.

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